ChronoTrace: Calibrating Residual Deleted Neighbor Inference After Node Deletion in Dynamic Trust Graphs

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Abstract

Deleting a node from a dynamic trust graph removes that node's incident records, but correlated structure in the retained Trust and reputation platforms publish who rated whom, and data-protection law obliges them to erase a user's data on request. Erasure deletes a user's ratings, yet the counterparties' other ratings stay public, so the published graph may still support inference of the deleted node's neighbor set. This paper presents reveal the user's financial and social relationships. Existing tools cannot measure this risk: link prediction needs the erased user's neighbors, unlearning certifies models rather than the published graph, and an attack's accuracy alone cannot separate the user's relationships from other users' popularity and timing. We propose ChronoTrace, an empirical benchmarkfor measuring deleted neighborinference after incident edge deletion in temporal signed trust graphs, evaluated on attack-and-calibration method, released as a benchmark. The attack ranks every remaining user as a likely former neighbor, with or without knowledge of when the erased user was active, and trains only on erasures it simulates on the published graph. The calibration reruns the attack on reference graphs that keep every user's degree, or signed activity over time, but randomize partners, and with degree-matched controls attributes its accuracy to degree, timing, or what remains. On 60 erasures in each of three timestamped trust graphs (BitcoinOTC, BitcoinAlpha, and Epinions(three public datasets, two domains; Epinions is a static snapshot to which we assign proxy record-order timestamps, so its temporal results are indicative only). The benchmark separates nonoracle attackers who observe only the released graph from oracle window attackers who additionally know the deleted node's interaction window. The central finding is that post-deletion degree alone achieves high AUC (0.838–0.896), but degree-proportional null model calibration reveals that this reflects heavy-tailed structural priors on all three datasets, with genuine deletion leakage beyond this structure only appearing on BitcoinOTC . This "negative" result is itself a privacy finding: platforms cannot assume that deleting a node's records removes its neighborhood information, because the structural prior already encodes neighbor likelihood. Oracle window temporal features provide the largest gain beyond this structural baseline, with the fused Epinions), the attack without timing knowledge reaches an area under the ROC curve (AUC) of 0.87–0.92 (0.5 is chance), but degree explains it: among users of equal degree it performs at chance (degree-matched AUC 0.46–0.52). On the Bitcoin trading graphs, a window covering the user's active months adds signal beyond degree (degree-matched AUC 0.81), and on BitcoinOTC sequential user IDs let an adversary estimate this window from the published graph alone, gaining +0.018 AUC, about half the exact-window gain. Precision stays low: top-five precision is at most 0.19. ChronoTrace attacker reaching AUC0.928–0.946, although on Epinions that gain over the best nonoracle feature set is not statistically distinguishable. Operational precision also remains low(P@5 of 0.10–0.21), meaning the attack narrows candidate pools rather than pinpointing specific neighbors. We position the work as an honest measurement benchmark where the null-model calibration methodology is a primary contributionthus lets platforms audit an erasure before publishing and identifies two cheap fixes: hiding accounts left without ratings and not numbering users in order of first activity.

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1 Introduction

A trust platform records relational facts: who trusted whom, when the judgment occurred, and whether the judgment was positive or negative. When a node is deleted, the natural operational response is to remove every record directly incident to that node. This satisfies a record centric view of deletion because the deleted node's own edges are gone. Relational data Trust and reputation platforms make relationships public by design. Traders on Bitcoin OTC, a marketplace for over-the-counter bitcoin trades, rate each other after a trade on a scale from −10 to +10 so that newcomers can judge a counterparty's history [17], and members of Epinions, a consumer review site, declared which reviewers they trusted and which they distrusted [22]. Data-protection law entitles each of these users to have their data erased: Article 17 of the EU General Data Protection Regulation requires a platform to erase a user's personal data on request [9], and a platform honors such a request by deleting the account and every rating the user gave or received.

Such erasure deletes the user's records but not the user's relationships, because every deleted rating had a counterparty whose other ratings stay public. Figure 1 shows a trader u who traded with users a, b and c. After erasure, these three users remain in the published graph with their other ratings; they were active in the same months as u, and, however, is not independent. Alice's trust

rating of Bob is an edge involving Alice, but Bob remains in the graph, Bob's other interactions remain visible, and Bob's structural position may still reveal that he was likely connected to Alice. The privacy question is therefore not whether the deleted records are still present. The question is whether the retained graph still carries enough correlated evidence to infer part of the deleted node's neighborhood. If Bob had 15 interactions in the week before Alice deleted her account, andthose interactions suddenly shift in timing or sign balance, an adversary observing the released graph can infer that Bob lost a neighbor around that time, and then use Bob's degree and neighborhood structure to rank which of the thousands of remaining nodes was most likely to be that neighbor. like most active traders, they are better connected than a typical user. An adversary who knows that u used the platform, such as a former counterparty, an investigator or a data broker, can therefore rank the remaining users by how likely each is to be one of u's *former neighbors*, the users that u rated or was rated by. Trading relationships are financial associations and trust relationships reveal social affiliations, so such a ranking is a privacy risk even when no single relationship is certain. The same risk arises for dataset custodians, who honor takedown requests by removing a user from a published graph [19, 24], and for machine-learning pipelines, whose exact unlearning retrains on the graph that remains [3, 4].

This paper studies that question as an empirical released graph inference problem. Given a temporal signed trust graphafter incident edge deletion, an attacker ranks every remaining node by the likelihood that it was formerly adjacent to the deleted node. The task is deliberately narrower than modelinversion, graph neural network edge privacy, or certified graph unlearning. We do not attack a trained neural graph model, and we do not prove a formal deletion guarantee. We measure residual leakage in the released graph itself. Existing tools leave this risk unmeasured. Link prediction [20] scores pairs of users who are both still in the graph, mostly through the neighbors they share. An erased user has no neighbors left, so every neighborhood-based score is zero, and preferential attachment, once the erased user's degree is set to a constant, ranks candidates by their own degree. Missing-node inference and cold-start link prediction [10, 18] reconstruct nodes or links that were never observed, whereas an erased user is known and was removed deliberately. Graph unlearning [5, 7] removes a user's influence from a trained model, which at best makes the model equivalent to one trained on the remaining graph; what that graph itself reveals lies outside the guarantee of unlearning.

The benchmark uses classical graph features extracted from the released adjacency matrix and temporal edge table, not GNN or autoencoder representations. This choice separates released-graph inference from model-level inference and enables transparent ablation across feature groups (Section 4.1). Even an attack that ranks former neighbors well would leave the risk unexplained, because

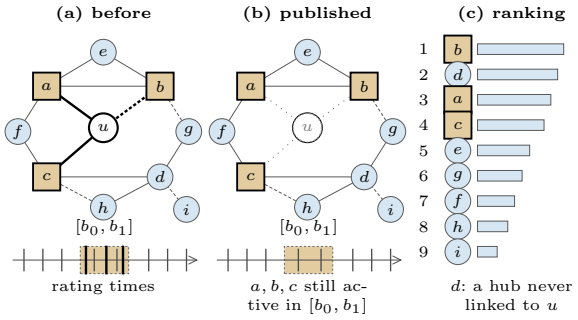


Figure 1: Deleting a node's records is not the same as deleting its relational traceRunning example. (a) A signed, timestamped graph around target Trader u ; the heavy edges E_u are the ones deleting u removestraded with a, and their time span$b$, binned into 32 uniform bins and padded by one bin on each side, is the deletion c (squares) in window $[b_0, b_1]$. (b) The released graphErasure deletes u's ratings. E_u is gone (dotted), yet the former neighbours $N_u = \{a, b, c\}$ keep residual degree 3, above the 2 or fewer of the ordinary nodes c to i, and their surviving edges still fire inside $[b_0, b_1]$. (c) Ranking the candidate set C_u by that residue lifts N_u, yet it also lifts the non-neighbour hub d, because degree alone already predicts adjacency. A degree-proportional null model prices that structural prior in, so we report ranking of the AUC gap above the null rather than the raw AUC (Table ??)remaining users. Scores and bar lengths are schematicSolid: positive rating; dashed: negative.

a high accuracy need not reveal anything specific to the erased user. Former neighbors tend to be well connected, so ranking users by degree already finds many of them, as it would in any graph in which the same users have the same degrees. Likewise, users who were active in the same months as the erased user are plausible partners in any graph with the same activity over time. An accuracy figure therefore cannot tell a platform whether its published graph exposes the erased user's relationships or only the popularity and timing of its users, yet the answer decides which remedy can work. What is missing is a method that measures this residual inference under explicitly stated adversary knowledge and attributes it to degree, to activity timing, and to what remains beyond them.

The central difficulty is observability. Some signals are available to a realistic attacker who only sees the released graph. Residual degree, signed balance, directional balance, local topology, and temporal concentration can all be computed after deletion. Other signals require privileged benchmark side information, especially the true historical interaction window of the deleted node. We therefore evaluate two clearly separated settings. The nonoracle setting measures what is inferable from the released graph

alone. The oracle window setting is an upper bound stress test that asks how much additional signal appears when the attacker This paper asks how much the published graph reveals about an erased user's former neighbors, and why. We propose ChronoTrace, an attack-and-calibration method, released as a benchmark. The attack ranks every remaining user under two separate levels of knowledge: a *release-only* adversary sees the published graph and the erased user's pseudonym, and a *window-informed* adversary also knows when the deleted node was active. This distinction prevents the strongest numbers from being mistaken for a realistic attacker unless the required side information is actually available.

We implement the benchmark as erased user was active, exactly or approximately, from side information such as old public posts or receipts of past trades. Either adversary trains on erasures that it simulates on the published graph itself, so it needs no ground truth about the erased user and never sees the user's own ratings. The calibration runs the same adversary on *reference graphs* that keep every user's degree, or every user's positive and negative activity in each period, and randomize who interacted with whom; two degree controls measure the accuracy that remains among candidates of equal degree. The gap between the real graph and each reference attributes the adversary's accuracy to degree, to activity timing, or to what remains beyond them. Unlike link prediction, ChronoTrace and evaluate it on three public temporal signed trust graphs: BitcoinOTC, BitcoinAlpha, and Epinions. For each target node, we remove all incident edges and evaluate deleted neighbor inference over the full candidate universe of thousands of remaining nodes. The supervised attacker is intentionally lightweight: logistic regression over hand engineered graph features with feature standardization and balanced class weights. It is classical supervised machine learning, not a neural or representation learning attack. We use leave one deleted node out training across deletion tasks and report metrics suited to extreme class imbalance, including AUC, AP, Recall@k, Precision@k, success@k, rank statistics, paired per task deltas, bootstrap confidence intervals, and deleted degree stratification. needs no neighborhood of the erased user; unlike graph unlearning, it examines the published graph rather than a model; and unlike an accuracy figure, it says where the accuracy comes from.

The central finding is that post-deletion degree is a strong nonoracle signal, but degree-proportional null model calibration reveals that most of this discriminative power reflects heavy-tailed degree structure rather than genuine deletion residues. This "negative" result is itself a privacy finding: it means that even without any deletion-specific signal, the structural prior already encodes neighbor likelihood. On BitcoinOTC, degree AUC exceeds the null by +0.041, confirming genuine leakage beyond structure; on BitcoinAlpha and Epinions, degree AUC falls below the null, indicating that the structural baseline alone suffices. Oracle windowtemporal features provide the largest gain beyond this structural baseline, with the fused We evaluate ChronoTrace attacker reaching AUC 0.928–0.946 (the Epinions value is computed under proxy record-order timestamps; Section ??); yet the gain is not uniform, because on Epinionson three signed trust graphs with genuine timestamps: BitcoinOTC, BitcoinAlpha and dated Epinions. Our contributions are as follows.

- **An erasure attack with explicit adversary knowledge** (Sections 3 and 4). Without timing knowledge it ranks former neighbors with AUC 0.87–0.92, the probability that a former neighbor is ranked above a non-neighbor (0.5 is chance); even with the exact window, the fused attacker is not statistically distinguishable from the best nonoracle feature set. Operational precision remains low (P@5 of 0.10–0.21), meaning the attack narrows candidate pools rather than pinpointing specific neighbors share of former neighbors among its top five candidates is at most 0.19.
- **A calibration that attributes the inference** (Section 4.4). Degree explains the inference without a time window: among candidates of equal degree the adversary performs at chance (degree-matched AUC 0.46–0.52). On the two Bitcoin trading graphs, the activity window adds signal beyond degree (degree-matched AUC 0.81) and ranks low-degree former neighbors above chance, as long as it covers the user's active months; on Epinions, whose trust statements are seldom reciprocated, the window does not improve the ranking.
- **Two practical side channels** (Sections 6.1 and 6.3). An account directory that keeps users without remaining ratings visible identifies those former neighbors exactly. In the BitcoinOTC release, user IDs follow the order of first activity and let a release-only adversary recover about half of the exact window's gain, +0.018 AUC over ranking by degree.

The published graph also indicates the sign and the approximate time of a deleted relationship, and the two simple defenses we test, local edge suppression and edge-level randomized response, leave the degree and timing signals largely intact. We release the code and every per-target result, from which our scripts regenerate every number, table and figure. Section 2 positions the work, Sections 3 to 5 define the problem, the benchmark and the setup, Section 6 answers five research questions, and Sections 7 and 8 discuss implications and limitations.

This paper makes three primary contributions. First, it introduces the empirical benchmark for deleted neighbor inference, evaluated across multiple domains with explicit nonoracle/oracle separation. Second, it introduces degree-proportional null model calibration as a methodology for separating genuine deletion leakage from structural artifacts, revealing that the heavy-tailed structural prior itself encodes neighbor likelihood. Third, it demonstrates that oracle window temporal features provide the largest gain beyond the

structural baseline, while providing an honest accounting of operational utility: practical precision remains low, meaning the attack narrows candidates rather than pinpoints specific neighbors.

2 Background and Related Work

We organize related workinto areas that motivate the problemand situate the contribution, without claiming that prior work directly studied deleted neighbor inference under our specific setup

Trust graphs and erasure in practice. A trust graph records ratings between users: who rated whom, when, and whether the rating was positive or negative. Reputation systems compute trust scores from such graphs to warn users about fraudulent counterparties [17], and review sites use them to filter content and recommendations [22]. Both endpoints of every edge are people, which distinguishes these graphs from bipartite user–item rating graphs, where one side consists of products and the relationship between two people is never recorded; we therefore study user–user graphs. Erasure from a user–item graph raises the analogous question for a user's purchases or reviews; the benchmark applies to it without change, but we do not evaluate it. The BitcoinOTC, BitcoinAlpha and Epinions graphs are public research releases [19, 24]. When a user leaves, a platform or custodian typically deletes the user's node with every incident edge, which is the deletion operator we study.

2.1 Signed trust networks and link prediction

Signed networks model positive and negative relationships. Leskovec et al. showed that signed social media links obey structural balance and status regularities . BitcoinOTC and BitcoinAlpha are signed trust networks from cryptocurrency trading communities; prior work used them for edge weight and trust sign prediction . Link prediction heuristics such as common neighbors,

Link prediction, cold starts and missing nodes. Link prediction ranks node pairs by proximity in an observed graph, using common neighbors, Jaccard similarity, Adamic Adar, and preferential attachment Adamic–Adar, Katz or preferential attachment [1, 20]form a standard toolbox for scoring link likelihood in static graphs. Our task is not ordinary link prediction: the deleted endpoint is absent. We adapt these heuristics as retained prominence scores conditioned on temporally active anchors, and we distinguish between heuristics that require oracle window anchor information (common neighbors, Jaccard, Adamic Adar, preferential attachment) and those that do not (degree prior). The adapted heuristics are task specific residualizations rather than community standard baselines for this exact task.

2.1 Temporal graph structure and graph privacy attacks

Temporal networks encode interaction timing as well as topology . In a deletion setting, the target node's historical interaction window matters because retained nodes that were temporally coactive with the deleted node may exhibit residual burst or shift patterns. We assume the deleted node is known and test whether the remaining temporal signed graph supports inference of that node's neighbor set; Section ?? contrasts this with the re-identification literature.

2.1 Deletion, unlearning, and residual inference

Machine unlearning studies how to remove the influence of deleted training data from learned models . Our work is complementary but distinct. We do not implement a formal unlearning algorithm or evaluate model level influence removal. We study a different residual risk: after direct record deletion, the retained relational records of other nodes may still encode correlated evidence about the deleted node's historical interactions. This residual risk sits between the record erasure guarantee of deletion and the model-level influence removal of unlearning, and it motivates the empirical measurement approach we take. These scores need the neighborhoods of both endpoints, which an erased user lacks, so after erasure they reduce to zero or to a ranking by degree (Section 1). Section 6.2 shows through calibration that features which need no time window add nothing to this ranking; what lies beyond it comes from timing knowledge and from side channels. Cold-start link prediction predicts links for nodes without network data from side information such as group memberships [18]. Missing-node identification and network completion infer nodes and edges absent from an observed graph [10, 16]. Our setting differs from these in three ways: the deleted user is known and its absence is deliberate; the question is which retained users were its neighbors; and every accuracy is calibrated against reference graphs that preserve degree or activity, which tells a platform whether the published graph reveals anything beyond those properties.

2.1 Modern graph privacy and unlearning

The graphprivacy and unlearning landscape has advanced substantially. GraphEraser introduced partition-based graph unlearning via sharded retraining, establishing evaluation standards (utility retention, membership inference, runtime) that inform our multimetric reporting. Chien et al. study certified graph unlearning with formal removal guarantees, sharing our deletion concern but targeting model level certified removal rather than released graph residual inference. Wu et al. propose influence function based graph unlearning, GUIDE extends graph unlearning to inductive settings, and Unlink to Unlearn simplifies edge unlearning in GNNs. Said et al. survey graph unlearning paradigms

including exact, approximate, and certified methods. Cheng et al. propose GNNDelete, a general layer-wise deletion operator for GNN unlearning.

On graph privacy attacks,

Privacy attacks on graph models and graph unlearning. LinkTeller [32]recovers private , GNNBleed [27] and black-box edge attacks [14] recover edges from a trained GNN via influence analysis of model query responses, establishing edge-level graph privacy as an active concern, though its threat model (model queries) differs from ours (raw released graph analysis). Song et al. present GNNBleed, which infers private edges from GNN modelsunder realistic access assumptions, advancing edge-level graph privacy attacks with a threat model based on model queries rather than released graph analysis. He et al. study black-box edge privacy attacks on GNNs, demonstrating that edge-level information can be extracted without full model access. He et al. study node level membership inference against GNNs. Wang et al. graph neural network, and further work infers node membership [15], subgraph structure [31] investigate subgraph structure membership inference attacks against GNNs, showing that structural patterns leak membership information. Zhang or graph properties [35] from trained models. Graph unlearning removes deleted nodes or edges from trained models, by sharded retraining [5], with certified guarantees [7], through influence functions [33], in inductive settings [30], or through layer-wise deletion operators [6, 28]; Said et al. survey the area [25]. All of these, like deletion inference present a systematic study of inference attacks against GNNs, spanning property inference, subgraph inference, and graph reconstruction. Olatunji et al. study differential privacy guarantees for released GNNs, providing a formal privacy framework complementary to our empirical measurement approach.

Common to all of these is a [11], treat a trained model as the attack surface or the object of removal; we attack neither, and study instead what the retained graph itselfreveals. We study the published graph itself, which bounds what any model trained on it after exact unlearning can still encode.

Our work also differs from the structural

De-anonymization. Structural re-identification literature. Backstrom et al. mount active and passive attacks to re-identify nodes across links anonymized graph releases ; Narayanan and Shmatikov de-anonymize social networks by cross-referencing auxiliary graphs ; Hay et al. study resistance to structural re-identification. These works share our concern that graph structure leaks private information, but they target *identity recovery* across graph releases, whereas we target *neighbor membership inference* for a known deleted node in a single released graphto identities through auxiliary graphs or planted structures [2, 13, 23]. Our adversary already knows the erased user's identity and asks which retained users were its neighbors. Table 1 summarizes this positioning. No prior work measures what the retained graph topology reveals about a deleted node's neighborhood after incident edge deletion in temporal signed trust graphs, using calibrated null models to separate genuine leakage from structural artifacts. This is the gap ChronoTrace fills.

3 Problem Statement and Formalism

3.1 Data model

Let A trust graph $G = (V, E)$ be a temporal signed directed graph . Each edge consists of users V and rating events E. Each event is

$$e = (s, d, \sigma, t), \tag{1}$$

where user s is the source node, rated user d is the destination node, at time t with sign $\sigma \in \{-1, +1\}$is the binarized trust sign (original ratings > 0 mapped ; ratings above zero map to $+1$, ratings ≤ 0 mapped to -1), and t is a timestamp. In our datasets , vertices are treated uniformly as graph nodes.

3.2 Deletion operator

For a target node $u \in V$, the set of incident edges is

$$E_u = \{(s, d, \sigma, t) \in E : s = u \vee d = u\}.$$

After deletion, the released graph is

$$G_u^{\text{rel}} = (V \setminus \{u\}, E \setminus E_u).$$

The candidate universe for reconstruction is $C_u = V \setminus \{u\}$, i.e., all nodes other than the deleted node.

3.2 Sensitive target

The sensitive quantity is deleted neighbor membership, defined as

$$N_u = \{v \in V \setminus \{u\} : \exists (s, d, \sigma, t) \in E_u,\ v = s \vee v = d\}.$$

N_u is the set of nodes that had each ordered pair rates at most once and ratings are never withdrawn, so an edge is a single timestamped event, and no assumption about how long a relationship lasts is needed. Two users are *neighbors* if at least one incident edge with u, ignoring edge direction, sign, and timestamp. The current benchmark evaluates membership of v in N_u; it does not evaluate which specific edges existed, their direction, their signs, or their timestamps. event links them in either direction. A user's *degree* is its number of events, given or received, and its *neighbor count* $|N(v)|$ is its number of distinct neighbors. To summarize activity, we divide each dataset's time span into 32 equal *time bins*, about two months on the Bitcoin graphs and one month on Epinions; the bins serve only to compute features.

3.2 Attack task

An attacker observes G_u^{rel} and knows the identity of u. The attacker may additionally have side information A_u about u's historical interaction window. The attacker outputs a score function $\text{score}(G_u^{\text{rel}}, u, v, A_u)$ for every candidate $v \in C_u$. The evaluation metrics are AUC and

Table 1: Positioning against related work.

Work	Infers	Attack surface	Timestamps	Signs	Target node present	Calibration
Liben-Nowell and Kleinberg [20]	future links	observed graph	split only	no	yes	none
Leroy et al. [18]	links of a node without edges	side information	no	no	no edges	none
Eyal et al. [10], Kim and Leskovec [16]	missing nodes and edges	observed graph	no	no	unknown	none
Backstrom et al. [2], Hay et al. [13]	node identity	released graph	no	no	yes	none
Narayanan and Shmatikov [23]	node identity	auxiliary graph	no	no	yes	none
LinkTeller [32], Maui [14]	training edges	GNN queries	no	no	yes	none
GNNBleed [27]	private edges	GNN access	yes	no	yes	none
Zhang et al. [35]	properties, subgraphs	GNN access	no	no	yes	none
GraphEraser [5], Chien et al. [7]	(removes model influence)	model parameters	no	no	removed	–
This work	**former neighbors, sign, time**	**published graph**	**yes**	**yes**	**erased**	**degree, activity**

~~AP (threshold independent), supplemented by Recall@k for $k \in \{50, 100, 500\}$.~~

4 ~~Benchmark Design~~

~~The pipeline for one deletion task, with the degree-proportional null model that calibrates it. The platform drops every edge incident on target u, optionally applies to what remains, and publishes the released graph G', so edits G' before any feature is built and is a defense rather than a step of the attack. The eleven cells are the coordinates of the per-pair feature vector $\mathbf{x}_{u,v}$, defined in Section 4.1 and mapped to ablation groups in Table 2. The shaded, dashed cells 3 to 6 form the contiguous oracle-window block $\mathcal{O}$; the other seven form the nonoracle block $\mathcal{N}$. The vector is a function of the pair and not of v alone, because deleting E_u lowers the released degree of exactly the neighbours of u, and because the $\mathcal{O}$ cells index the temporal histogram of v by the window of u. Scores come from a standardized, class-balanced logistic model fitted leave-one-deletion-out (LODO), that is, on the other deletion tasks only, with negatives pooled across them. The null branch consumes the same released graph and the same candidate set, so the difference between the two AUCs isolates the leakage that the degree structure of G' does not already explain.~~

~~One requirement shapes every choice in this benchmark: separating genuine deletion-specific leakage from structural artifacts that would exist whether or not any node had been deleted. The design rationale, architecture, and evaluation framework below all follow from it.~~

3.1 ~~Benchmark architecture and observability branching~~

~~Every quantity the benchmark computes is tagged with the attacker knowledge it requires: nonoracle (N), computable from the released graph alone; oracle window (O), requiring the deleted interaction window $[b_0, b_1]$; and mixed (M), combining the two. Section 3.1 defines the corresponding attackers and Table 2 assigns every feature, ablation group, and heuristic to one category. Tagging is what keeps the strongest numbers from being read as a realistic attack: no result in this paper appears without the knowledge assumption that produced it.~~

3.1 ~~Task construction and sampling protocol~~

~~Each benchmark task isolates a single deleted node~~

3.1 Erasure and the published graph

Erasing user u ~~and measures whether the remaining graph supports inference of u's deleted neighbor set N_u. Target selection samples nodes across degree buckets (minimum incident degree 4, maximum 128) to capture nontrivial neighborhood sizes while excluding extreme outliers. The lower bound of 4 ensures that the reconstruction task is nontrivial (nodes with degree 1 3 have trivially small candidate neighborhoods whose membership is easy to guess or uninteresting to protect). The upper bound of 128 excludes extreme hub nodes whose deletion would structurally dominate the graph; such nodes constitute fewer than 2% of nodes in all three datasetsand would inflate aggregate metrics. The resulting 60 tasks per dataset (Section ??) support aggregate means and bootstrap confidence intervals, yet they limit formal significance testing of small inter-method differences, so we do not report differences below approximately 0.03 AUC as statistically robust.~~

~~The candidate set is $C_u \subseteq V \setminus \{u\}$, capped for tractability, and Appendix ?? shows the cap does not materially affect AUC (mean shift <0.01 across 12 tasks tested at 2~~deletes every event in which u is the rater or the rated user,

$$E_u = \{(s, d, \sigma, t) \in E : \; s = u \;\vee\; d = u\}, \tag{2}$$

and the platform publishes

$$G_u^{\mathrm{rel}} = (V \setminus \{u\},\; E \setminus E_u), \tag{3}$$

which we call the *published graph*. Publication takes one of two forms. An *edge-list release*, the format of public research datasets, ~~000 and 10,000 candidates). Extreme class imbalance is inherent: each deleted node has only a handful of true neighbors among thousands of candidates, which is why we lead with threshold-independent metrics.~~ lists events only, so a user left without events disappears. A *directory release*, such as a live platform's list of accounts, still lists every user. The distinction matters: in a directory release any user left without events must have interacted only with u, so those former neighbors are identified exactly. Section 6.1 measures this channel; every other result uses

edge-list releases, so those results measure inference beyond this lookup. The *candidates* C_u are the users present in the published graph.

3.2 Feature abstraction and ablation design

The supervised attacker uses 11 engineered features, and the design evaluates them in nested groups rather than all at once, so that each increment of attacker knowledge is priced separately: three nonoracle groups isolate what residual structural prominence alone yields, and the mixed groups then measure what the deletion window adds on top. The groups are asymmetric by construction, since nonoracle features remain available once oracle features are added but not the reverse, which is what makes the increments interpretable as a knowledge hierarchy. Section 4.1 lists the exact feature indices in each group.

3.2 Evaluation framework and significance methodologySensitive target

The protected information is the set of u's *former neighbors*,

$$N_u = \{v \in V \setminus \{u\} : \exists (s, d, \sigma, t) \in E_u \text{ with } v \in \{s, d\}\}. \quad (4)$$

Membership in N_u ignores direction, sign and time. Section 6.4 adds two secondary targets for each identified former neighbor: whether the deleted relationship included a negative rating, and when it happened.

Three commitments govern how results are read. AUC leads because it is threshold independent and robust under extreme imbalance, with AP, P@k, R@k, success@k and rank statistics (Section **??**) carrying the operational picture that AUC hides. Every pairwise comparison carries a bootstrap 95% confidence interval over per-task values, so task-level variance is visible without assuming normality. Above all, no nonoracle AUC is interpreted on its own: each is read against the degree-proportional null (Section **??**), which preserves the degree distribution the attacker sees while destroying actual adjacency, so that heavy-tailed structure is priced in before any leakage is claimed. Defenses are additionally charged for their cost, in edge retention and in sign-prediction AUC on heldout edges.

3.3 Design choices and scope boundaries

Three choices bound what the benchmark can claim. It ranks with logistic regression over hand-engineered features rather than a graph neural network, buying interpretability and ablation clarity at some cost in peak performance, a trade the DGTEN comparison in Section **??** suggests is cheap here. It binarizes trust signs and deletes one node at a time, leaving multi-node and multi-release attacks to future work. Finally itevaluates leave-one-deleted-node-out, so the adversary trains on other tasks' labels and never sees the target's deleted edges, modelling an attacker who has studied historical releases with known deletions but faces a fresh one.

4 Threat Model and Side Information Assumptions

3.1 Attack task and threat model

We distinguish two attacker knowledge settings. All results are labeled accordingly.

Oracle window attacker. The attacker knows the true earliest and latest timestamps among The adversary knows u's pseudonym, observes G_u^{rel}, and may hold side information A_u about when u was active. It outputs a score

$$\text{score}(G_u^{\text{rel}}, u, v, A_u) \quad \text{for every } v \in C_u, \quad (5)$$

and ranks the candidates by it. We separate two levels of knowledge, and every result states which level produced it.

- A *release-only* adversary has $A_u = \emptyset$: it reads only G_u^{rel} and u's pseudonym. It can rank candidates by features that need no time window, or estimate u's activity window from G_u^{rel} (Section 4.3).
- A *window-informed* adversary also knows u's deleted incident edges. These timestamps are mapped to time bins and then activity window $W_u = [b_0, b_1]$, the time bins spanned by the events in E_u, padded by one bin on each side, which defines the temporal window $[b_0, b_1]$ used by features 3 (window activity ratio), 4 (pre/post shift), 5 (positive window ratio), and 6 (negative window ratio). The padding makes the window robust to edges falling near a bin boundary; it widens the oracle window slightly and therefore, if anything, weakens the oracle attacker. It also defines the anchor set for adapted heuristic baselines (common neighbors, Jaccard, Adamic Adar, preferential attachment, temporal burst/shift, signed motif). This setting is . We evaluate the exact window, an upper bound stress test, not a model of a realistic released graph only attacker.

Restricted nonoracle attacker. The attacker on timing knowledge, and the exact window shifted by δ bins in a direction the adversary does not knowthe deleted interaction window. Available features are limited to what can be computed purely from G_u^{rel}: features 0 (log degree), 1 (signed balance), 2 (directional balance), 7 (log neighbor degree), 8 (log two hop reach), 9 (clustering proxy), and 10 (temporal concentration, an entropy based measure of overall burstiness that does not depend on any target specific window). Three nonoracle supervised ablations are evaluated, spanning increasing feature sets: degree (feature 0 only), topology (features 0, 7, 8, 9), and full nonoracle (features 0, 1, 2, 7, 8, 9, 10). Among the adapted heuristics, degree prior is the only nonoracle baseline; all other adapted heuristics use oracle window anchor sets or burst/shift terms and therefore belong to the oracle window setting.

Training assumption. The attacker can train on other deletion tasks sampled from .

Figure 1 instantiates these definitions: $N_u = \{a, b, c\}$, W_u is the shaded band, and the ~~same dataset, using leave-one-deleted-node-out evaluation. This models an adversary who can simulate deletion on public or historical graph releases but does not know the target's deleted labels.~~

~~**Out of scope.** The benchmark does not evaluate recovery of deleted~~ candidates are a to i. The threat model also excludes stronger adversaries. The adversary holds no copy of the graph from before the erasure, because comparing two snapshots would reveal N_u directly, and it attacks one erasure at a time. Recovering edge directions, ~~signs, or timestamps. It does not model attacks on raw storage, cryptographic attacks, or model inversion against~~ attacking storage, and inverting a separately trained ~~neural trust predictor~~model are out of scope.

~~**Estimated-window attacker.** A natural intermediate between the nonoracle and oracle settings is an attacker who *estimates* the deletion window from observable graph features, such as temporal gaps, burst anomalies, or degree-change patterns in the retained graph. Our implementation includes a simple deterministic estimator of this kind (the median timestamp of edges incident to the top-degree retained nodes, expanded to a fixed-width bin window), and it will form part of the release described in the Open Science section; we do not report its quantitative evaluation in this paper and defer a systematic study of estimation quality to future work. Reliably estimating the deletion window from the post-deletion graph in general requires a learned detection model whose accuracy would depend heavily on the graph's temporal dynamics and deletion prevalence. Consequently, the realistic released-graph attack falls somewhere between our nonoracle baseline (which ignores temporal anomalies)and the oracle window upper bound. Future work should evaluate learned estimated-window attackers with varying estimation quality.~~

3.2 Metrics

A ranking is scored over all candidates. AUC, the area under the ROC curve, is the probability that a randomly chosen former neighbor ranks above a randomly chosen non-neighbor, so 0.5 is chance. AP (average precision) averages the precision at the rank of each former neighbor. P@k is the fraction of the top k candidates that are former neighbors, R@k the fraction of former neighbors in the top k, and S@k is 1 if at least one former neighbor is in the top k; the best and mean ranks locate the former neighbors in the list. Two further metrics isolate what degree cannot explain. The *degree-matched AUC* compares only former neighbors and non-neighbors with the same number of events in G_u^{rel}, so ranking by degree scores exactly 0.5. The *excess over degree* is the AUC minus the AUC expected if the former neighbors were redrawn at random among candidates with the same degrees (Section 4.4).

4 ~~ChronoTrace Method~~

3.1 ~~Overview~~

3.1 Difficulty

Exact recovery by guessing is hopeless for most users. Between 64% and 79% of the users in our datasets have at most three neighbors, and guessing even a three-member N_u exactly at random among 5,880 BitcoinOTC candidates succeeds with probability below 10^{-10}. A ranking is therefore the meaningful output, and we report its quality at every depth.

~~Having established the threat model and observability categories, we now describe how the benchmark pipeline implements them.~~

4 The ChronoTraceBenchmark

Figure 2 ~~summarizes the undefended attack path of the implemented pipeline. For each target node~~ shows the method on the running example. The platform deletes u's events, optionally applies a defense, and publishes the result. The adversary trains a model on erasures it simulates itself (Figure 2a), describes every candidate by features of the published graph, adds timing features if it holds a window, ~~the system removes all incident edges, derives the deletion window from those edges, optionally applies the scrubbing transformation to the remaining edges, constructs features for every candidate from the resulting released graph, trains a logistic-regression attacker on held-out deletion tasks, and evaluates the held-out task. Two elements are omitted from the figure for legibility and described in the text instead: , which when enabled transforms the released graph before feature extraction rather than after it (Section ??), and the degree-proportional null model, which is computed per target alongside the attacker and against which every nonoracle result is calibrated (Section ??).~~

4.1 ~~Observability matrix~~

~~Features 0,1,2,7,8,9,10 are nonoracle (N); features 3, 4,5,6 are oracle window(O). Supervised ablations: degree, topology, and full nonoracle are N; temporal, signed-structural, and chronotrace are mixed (M). Among heuristic baselines, only degree prior is N; all others (common neighbors, Jaccard, Adamic-Adar, preferential attachment, temporal burst/shift, signed motif, CT-unsupervised)are O~~ranks the candidates, and is scored against N_u (Figure 2b). The same attack runs on reference graphs to calibrate the result (Figure 2c).

4.1 ~~Feature extraction and ablation groups~~

~~The implementation constructs an 11-dimensional feature vector per candidate node.~~

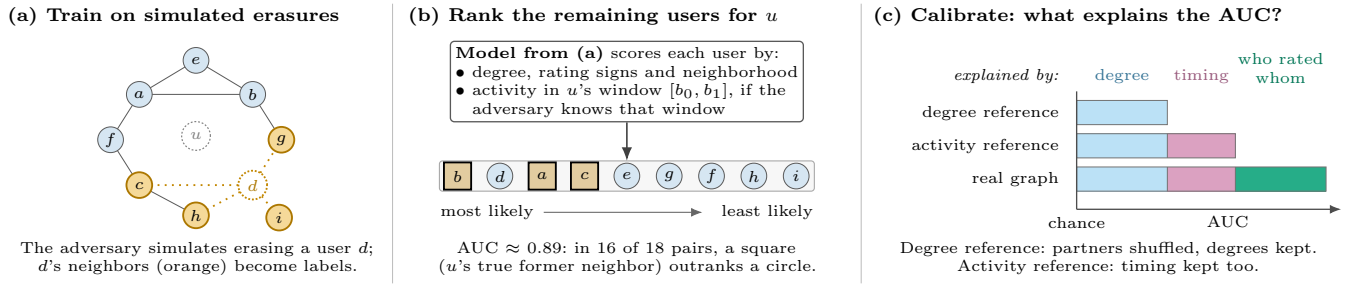


Figure 2: How ChronoTrace works on the example of Figure 1. The platform erased u. The adversary erases d only in simulation and never sees u's former neighbors; only the AUC uses them. In (c), the whole attack, training included, reruns on each graph; bar lengths are illustrative.

4.1 What the published graph reveals

ChronoTrace describes each candidate v by eleven features computed on G_u^{rel} (Table 2 ~~lists every feature with its index, observability category, and the supervised ablation groups in which it appears. All reported ablation groups correspond exactly to the feature indices listed here.~~

~~*Feature definitions.* The 11 engineered features $x_{u,v}$ computed for each candidate~~ ; exact definitions in Appendix A). Popularity is captured by v ~~given target u are defined as follows, and match the released implementation exactly. Write p_v and n_v for the number of positive and negative edges incident to v in G_u^{rel}, and o_v and i_v for its outgoing and incoming edge counts.~~ ('s degree (feature 0) ~~*Log degree:* $\log(1+\deg(v))$ in G_u^{rel}. (1) *Signed balance:* $(p_v - n_v)/\max(1, p_v + n_v)$, the normalized excess~~ . Signs and directions enter through v's balance of positive over negative ~~incident edges, in $[-1,1]$. (2) *Directional balance:* $(o_v - i_v)/\max(1, o_v + i_v)$, the normalized excess~~ events and of outgoing over incoming ~~edges. (3) *Window activity ratio:* the fraction~~ events (features 1 and 2). Local structure enters through the mean degree of v's ~~total edges that occur within the oracle deletion window $[b_0, b_1]$. (4) *Pre/post activity shift:* $|\bar{h}_{\text{post}} - \bar{h}_{\text{pre}}|$, where $\bar{h}_{\text{pre}}$ and $\bar{h}_{\text{post}}$ are~~ neighbors, the number of users two hops away, and a closure ratio (features 7 to 9). Time enters in two ways. Feature 10 measures how concentrated v's ~~mean per-bin edge counts over the three bins immediately preceding b_0 and the three immediately following b_1; the difference is not normalized by degree. (5,~~ activity is across the 32 bins and needs no knowledge of u. Features 3 to 6 ~~) *Positive/Negative window ratio:* the fraction of~~ align v's ~~positive (or negative) edges that fall within $[b_0, b_1]$. (7) *Neighbor degree:* $\log\left(1 + \frac{1}{|N(v)|}\sum_{w\in N(v)} \deg(w)\right)$, the log mean degree~~ activity with the window W_u: the share of v's ~~neighbors (zero when $N(v) = \emptyset$). (8) *Two-hop reach:* $\log(1+|T(v)|)$, where $T(v)$ is the set of nodes reachable in exactly two hops from~~ events inside the window, the change in v, ~~excluding v itself and its direct neighbors. (9) *Clustering proxy:* $\deg(v)/\sqrt{1+|T(v)|}$, a degree-to-two-hop-expansion ratio that is large when v's neighborhood closes back on itself. It is a cheap surrogate for neighborhood closure and is *not* the local clustering coefficient. (10) *Temporal concentration:* $1 - H(v)$, where $H(v)$ is the Shannon entropy~~ 's activity across the window boundaries, and the shares of v's ~~edge timestamps discretized into 32 uniform bins and normalized to $[0,1]$, so that higher values indicate burstier activity. All features are standard-scaled prior to classification.~~ positive and of its negative events inside the window. Features 3 to 6 need a window: the exact or shifted window of a window-informed adversary, or a window that a release-only adversary estimates from G_u^{rel}.

~~*The temporal ablation group includes oracle features3 and 4 alongside nonoracle feature 10. The signed-structural ablation group includes nonoracle features 1, 2~~

4.2 Adversaries and how they train

Each adversary scores candidates with a logistic regression on standardized features, ~~7,8,9 alongside oracle features 5,6. These groups are labeled as mixed observability in all tables and prose.~~ with balanced class weights. We use a linear model so that its coefficients show which evidence drives the ranking (Section 6.2); Appendix D.9 shows that a random forest ranks no better. We name an adversary by its feature set (Table 2) and, for sets that need a window, by its window. The *window-free* sets are Degree, Topology and Full-R, which uses every feature that needs no window. The *window-based* sets are Temporal, Sgn-Struct (signed-structural) and ChronoTrace, which uses all eleven features; we name this full attacker after the benchmark.

Six ablation groups are evaluated: ~~**degree** (N, feature 0) , **topology** (N, features 0, 7, 8, 9) , **full nonoracle** (N, features0, 1, 2, 7, 8, 9, 10), **temporal** (M, features 3, 4, 10), **signed-structural** (M, features 1, 2, 5, 6,7,8,9), and **chronotrace** (M, all 11 features). The deletion window $[b_0, b_1]$ is derived from the target's deleted edges (oracle information)~~ An adversary needs labeled examples, and it can make them from G_u^{rel} alone. To attack the erasure of u, it picks other users $u_1, \ldots, u_m$, erases each from G_u^{rel} itself, and labels each simulated erasure with the neighbors that u_j has in G_u^{rel}. These simulated *shadow erasures*, like the shadow models of membership inference [26], contain no event of u and no label that mentions u, so nothing about the target

Table 2: Candidate features, the information each reads, and the feature sets that use it.

			Window-free			Window-based		
#	Feature	Reads	Degree	Topology	Full-R	Temporal	Sgn-Struct	ChronoTrace
0	Degree	degree	✓	✓	✓			✓
1	Sign balance	sign			✓		✓	✓
2	Direction balance	direction			✓		✓	✓
3	Window activity share	time, W_u				✓		✓
4	Activity shift at window	time, W_u				✓		✓
5	Positive window share	time, sign, W_u					✓	✓
6	Negative window share	time, sign, W_u					✓	✓
7	Neighbor degree	structure		✓	✓		✓	✓
8	Two-hop reach	structure		✓	✓		✓	✓
9	Closure ratio	structure		✓	✓		✓	✓
10	Activity concentration	time			✓	✓		✓

reaches training. The adversary pools the shadow erasures, keeps every former neighbor and max(1000, 20 × positives) non-neighbors, and fits the model. Any users present in G_v^{rel} can serve as shadow users; we use the other eleven targets drawn with the same seed, so that the two training protocols compared in Section 6.1 train on the same users.

4.3 Timing knowledge

The exact window W_u is the benchmark's upper bound on timing knowledge. Because W_u is padded by one bin, a window shifted by one bin still covers every bin in which u was active, and larger shifts miss some of them. To model imperfect knowledge, we shift W_u by $\delta \in \{1, 2, 4, 8\}$ bins in a random direction, keeping its width, and shift the window of every shadow erasure by the same δ in an independently drawn direction, so that training sees errors of the same size as the test. Two estimators use only G_v^{rel} and u's pseudonym. The *hub* estimator centers a seven-bin window on the median time of the published events of the 50 users with the most neighbors; it depends on the target only through the deleted events and in our data gives every target of a dataset the same window. The *ID-cohort* estimator exploits pseudonyms numbered in order of first activity: it takes the 25 users whose IDs are closest to u's and returns the median of their first active bins and of their last active bins, padded by one bin. We report the intersection over union (IoU) of every estimate with W_u.

4.4 ~~Supervised reconstruction~~Calibration

An AUC alone measures risk but not its cause, and the cause decides which defense can work. ChronoTracetherefore compares each adversary's AUC on the real graph with its AUC on *reference graphs* that keep some properties of the real graph and randomize the rest [12, 21].

- The *degree reference* rewires the graph by permuting the rated end of every event at random. Every user keeps its number of events given and received, and the partners become random. Its *label reference* draws $|N_u|$ pseudo-neighbors from the real candidates with probability proportional to degree, which approximates u's neighbors under the rewiring.
- The *activity reference* permutes the rated end only among events in the same time bin with the same sign. Every user then keeps its number of positive and of negative events given and received in every bin, so its degree and every timing feature are unchanged, and only who interacted with whom is randomized.

We erase the same targets in each reference graph and run the same adversary. The *gap* to a reference, the real AUC minus the reference AUC, measures how much better the adversary ranks the true former neighbors than neighbors that are random given the preserved properties. The deletion is the same in both graphs, so a gap reflects how partners were chosen in the original graph beyond the preserved properties. It can come from the reciprocity of ratings, from community structure or from how our targets were selected, as well as from the erased user's own relationships.

~~The supervised attacker uses logistic regression with feature standardization and balanced class weights. This is a lightweight classical supervised classifier over hand engineered graph features, not a neural graph model or embedding attack. We choose logistic regression for interpretability, ablation clarity, and scalability under extreme imbalance. Appendix D.9 confirms that random forest (100 estimators, max depth 10) consistently underperforms, with AUC dropping from 0.896 to 0.753 for degree-only features on BitcoinOTC, suggesting that tree-based models require more per-leaf samples than this imbalance provides. For a held out target node u_i, all other deletion tasks $\{u_j : j \neq i\}$ serve as training data. Because the candidate universe has extreme class imbalance, the training procedure pools the candidate rows of all training tasks into a single design matrix, keeps every positive instance, and draws one negative sample of size $\max(1000, 20 \cdot |\text{positives}|)$ from the pooled negatives, where $|\text{positives}|$~~ Two *degree controls* complement the rewiring. The degree-matched AUC restricts the comparison to candidates of equal degree. The *excess over degree* is the AUC minus the AUC expected when the former neighbors are redrawn at random within each degree class, which keeps their exact degree histogram. By linearity of expectation, the expected AUC under this redraw equals $(\sum_b p_b \bar{r}_b - |N_u|(|N_u|+1)/2)/(|N_u||C_u \setminus N_u|)$, where p_b

counts former neighbors of degree b and $\bar{r}_b$ is the ~~pooled positive count rather than a per-task count. AUC and AP are then computed over the full unsampled candidate universe. AUC is threshold independent and less sensitive to training classbalance; AP and Recall@k can be affected by the prevalence ratio and are best interpreted relative to the true positive count per task~~mean rank of degree-b candidates; in this formula only, ranks count upward from 1 for the lowest score.

4.5 Adapted heuristic baselines

~~Since the deleted node u is absent from G_u^{rel}, standard link prediction heuristics that use both endpointscannot be applied directly. The benchmark adapts them as retained prominence scores . An anchor set is defined as the most temporally active nodes in the target deletion window(oracle window). Its size is $\max(5, \min(50, 3|N_u|))$, so the anchor construction additionally consumes the true neighbor count $|N_u|$; this is oracle side information beyond the window itself, and it is one more reason the adapted heuristics are upper bounds rather than realistic attacks. Scores such as common~~ Classical link-prediction scores need both endpoints, and the erased user has no edges left. We therefore adapt them into scores of each candidate against an *anchor set*: the candidates most active inside the exact window, with $\max(5, \min(50, 3|N_u|))$ anchors. The anchor set reads the exact window and $|N_u|$, so every anchor-based heuristic is optimistic: it uses knowledge that a realistic attack lacks. Common neighbors, Jaccard ~~, Adamic Adar, and preferential attachment are computed between each~~ and Adamic–Adar compare a candidate's ~~residual neighborhood and this oracle window anchor set. Among the adapted heuristics, only degree prior $(\log(1 + \deg(v)))$ is a pure nonoracle baseline; all others are oracle window conditioned.~~ neighborhood with the anchor set; the *anchor overlap* of a candidate, its number of neighbors among the anchors, is its common-neighbors score. Preferential attachment multiplies the candidate's neighbor count by the anchors' mean neighbor count. The latter is constant within a target, so preferential attachment ranks exactly like the neighbor-count prior $\log(1 + |N(v)|)$. The window activity share and activity shift heuristics are features 3 and 4. Signed motif adds the absolute sign balance, the positive and negative window shares, and $\log(1 + \text{anchor overlap})$. *CT-Unsupervised* is a fixed, untrained combination of these signals: $\log(1 + |N(v)|) + \log(1 + \text{anchor overlap}) + 2 \times \text{window activity share} + \text{activity shift} + \text{signed motif}$.

4.6 ~~ChronoScrub defense baseline~~

~~While acts as the attack benchmark, we also introduce as a platform-side edge suppression stress test (not an attacker-side operation). The defender uses internal predeletion knowledge: which nodes were deleted neighbors and the true deletion window. identifies residue nodes (neighbors of deleted neighbors within a configurable hop radius) and suppresses edges through a deterministic hash-based decision rule: each edge is assigned a hash-derived keep threshold based on its endpoint IDs, sign, and timestamp, and is dropped if this threshold falls below a task-specific drop scorecomputed from the edge's structural and temporal position. The hash function uses a linear combination of endpoint IDs, timestamp, sign ,~~

4.6 Defenses

We evaluate two defenses a platform could apply before publishing. ChronoScrubis a diagnostic edge suppression that uses what the platform knows at deletion time, N_u and ~~a large prime multiplier, then takes the result modulo a fixedconstant. This is a deterministic, reproducible computation given the edge's attributes and does not involve random sampling. The temporal variant applies this rule to edges touching residue nodes inside the deletion window ; the joint variant additionally applies it to edges between pairs of residue nodes~~W_u. It marks the former neighbors and their neighbors as residue users. Its temporal variant deletes 45% of the events inside the window that touch a residue user, chosen by a deterministic hash of each event. Its joint variant at strength $s \in \{0.25, 0.50, 0.75\}$ deletes a share $0.45s$ of those events and also a share $0.25s$ of the events between two residue users. ChronoScrub ~~is provided only as a diagnostic baseline to measure whether simple temporal local scrubbing reduces the observed attack. It does not provide differential privacy, cryptographic security, or an indistinguishability guarantee.~~ offers no formal guarantee. Edge-level randomized response publishes each remaining event independently with probability $e^{\epsilon}/(1 + e^{\epsilon})$ for $\epsilon \in \{0.5, 1, 2, 4\}$; it only deletes events, so it falls short of edge-level differential privacy. The adversary knows the defense and simulates its erasures on the defended release; under ChronoScrubit also applies ChronoScrubto each simulated erasure, as the platform would. We also measure utility as the AUC of predicting the sign of the latest 20% of events from the defended release (Appendix B.6).

5 Experimental Setup

5.1 ~~Datasets~~

~~We now describe the experimental configuration that instantiates the benchmark design described above. The benchmark uses three real~~

Datasets. We use three signed trust graphs ~~spanning two domains. BitcoinOTC (5,881 nodes, 35,592 edges)~~with genuine timestamps (Table 3). BitcoinOTC and BitcoinAlpha record trader ratings on two over-the-counter bitcoin markets [17, 19]. Dated Epinions consists of the trust and ~~BitcoinAlpha (3,783 nodes, 24,186 edges) are cryptocurrency trust networks publicly distributed through the SNAP repository . Epinions (131,828 nodes, 841,372 edges) is a consumer review trust network from the Epinions.com platform, providing a non-cryptocurrency validation domain. All edges are directed, with binary trust signs derived from node ratings.~~ distrust statements of the Extended Epinions

release that carry a genuine creation date [22, 24]. The first recorded date of that release is a placeholder shared by 68.8% of all statements, and we drop those statements together with self-ratings. We number the dated users in order of their first rating, so the ID–time correlation of dated Epinions in Table 3 reflects this numbering rather than the platform's. The widely used static SNAP Epinions snapshot has no timestamps, only record order; Appendix E evaluates it as well.

Timestamps and the Epinions proxy time axis. BitcoinOTC and BitcoinAlpha carry genuine Unix timestamps for every edge. The public Epinions signed graph, in contrast, is a *static* snapshot with no native timestamps. To run the full pipeline uniformly, we assign each Epinions edge a *proxy timestamp* equal to its record order in the released edge list. This proxy axis is monotone in the file'srecord order (which groups edges by source node), so temporal and oracle-window quantities on Epinions measure *record-order locality*, not genuine interaction timing. Accordingly, we treat Epinions primarily as a *structural* validation domain: its degree, topology, and null-model results are timestamp-free and fully comparable, while its temporal and oracle-window numbers are reported for pipeline completeness only and are flagged as proxy-time results wherever they appear. All timing-based claims in this paper rest on the two Bitcoin datasets.

5.1 Target selection and protocol

For each seed and dataset,

Targets and protocol. For each of five seeds we draw 12 target nodes are sampled from degree buckets (degree 4–128), producing 60 tasks per datasetacross 5 seeds (42, 7users per dataset, 13, 100, 200). Each target is evaluated over the full candidate universe (all remaining nodes in the graph), yielding approximately 5,880 candidates for BitcoinOTC, approximately 3,782 for BitcoinAlpha, and up to 5, 000 subsampled candidates for Epinions (which has 131,828 nodes). Candidate subsampling preserves all true positives and randomly samples negatives to the configured maximum. The time axis is divided into 32 uniform bins. Primary metrics are AUC and AP (threshold-independent), supplemented by P@k, R@k, success@k, and rank statistics. All comparisons use bootstrap stratified over the quartiles of incident-event count between 4 and 128, which gives 60 erasures per dataset. The targets have between 2 and 99 distinct former neighbors (median 6), fewer than their events because many ratings are reciprocated. Section 6.4 adds 60 targets per dataset with one to three events, a group that contains 45% to 78% of all users. Every erasure is evaluated on its full candidate universe, without subsampling. Intervals are percentile-bootstrap 95% CIs confidence intervals over targets (10,000 resamples). Utility is measured by edge retention and sign prediction AUC. , differences are paired by target, and p-values come from two-sided sign-flip permutation tests with 10,000 random flips. Appendix B details the protocol. Comparisons between adversaries are Benjamini–Hochberg corrected within each dataset (Appendix D.2).

6 EvaluationResults

We organize results around four questions: (1) how much deleted neighbor signal is inferable from nonoracle structural features alone, (2) how much of that signal is genuine leakage versus a tautological consequence of degree distributions (null model calibration), (3) how much additional gain comes from oracle window temporal information, and (4) whether materially suppresses leakage. We address each question in turn, beginning with the supervised ablation results that establish the baseline leakage levelsthe results around five research questions, RQ1 to RQ5, each ending with an explicit answer; two subsections on heuristic baselines and validity checks follow.

6.1 Supervised ablation resultsRQ1: How much does the published graph reveal?

Table 4 reports deleted neighbor inference for the supervised ablations across all three datasets. The nonoracle degree ablation achieves AUC 0.896 ± 0.098 95% CI: 0.870, 0.919every adversary, and Figure 3 shows the same AUCs with their confidence intervals. The Degree adversary reaches AUC 0.910 on BitcoinOTC, 0.838 ± 0.145 0.801, 0.873on BitcoinAlpha , and 0.868 ± 0.112 0.837, 0.897on Epinions(5 seeds, 60 tasks per dataset). Topology adds +0.004 on OTC (CI includes zero), +0.033 on Alpha (CI excludes zero), 0.869 on BitcoinAlpha and +0.043 on Epinions (CI excludes zero). The full nonoracle set adds +0.004 on OTC (CI includes zero), +0.032 on Alpha (CI excludes zero), and +0.050 on Epinions (CI excludes zero). These paired CIs confirm that nonoracle features beyond degree provide statistically distinguishable improvement on BitcoinAlpha and Epinions but not on BitcoinOTC0.924 on Epinions.

The oracle window temporal and signed-structural ablations add substantial signal. The chronotrace vs. full nonoracle difference is +0.048 CI: +0.033, +0.064on OTC, +0.061 CI: +0.045, +0.077on Alpha, and +0.010 on Epinions (CI includes zero for the chronotrace vs. full nonoracle comparison, consistent with the high degree-proportional null (0.912) on Epinions, which leaves little room for improvement). Temporal oracle information therefore yields the largest gain on the two genuinely timestamped datasets; on Epinions, where the structural baseline is already high, the marginal gain over the full nonoracle set is not statistically distinguishable from zeroAdding signs, directions, structure and activity concentration does not improve on degree alone: neither Topology nor Full-R is significantly better than Degree on any dataset, and Full-R is slightly worse on BitcoinOTC (0.901; Appendix D.2). Figure 4 shows the per-target

Table 3: Datasets. Bin: days per time bin. Distrust: share of negative ratings. Recip.: share of ratings whose reverse exists; ≤7d: reversed within 7 days. ID ρ: Spearman correlation of user ID and first activity. ≤3 nbrs: users with at most three neighbors. Het.: degree heterogeneity $\overline{d^2}/\bar{d}^2$.

Dataset	Users	Ratings	Period	Bin	Distrust	Recip.	≤7d	ID ρ	≤3 nbrs	Het.
BitcoinOTC	5,881	35,592	2010-11 to 2016-01	59	10%	0.79	68%	1.00	65%	11
BitcoinAlpha	3,783	24,186	2010-11 to 2016-01	59	6%	0.83	70%	0.33	64%	8
Epinions	56,499	261,803	2001-01 to 2003-08	29	20%	0.29	18%	0.59	79%	30

Table 4: Former-neighbor ranking on edge-list releases (mean over 60 targets; AUC with 95% CI). Random: random scores. Window: none, estimated from the published graph, or exact. Features: indices from Table 2. DM-AUC: degree-matched AUC. Bold: highest AUC.

Dataset	Feature set	Window	Features	AUC [95% CI]	AP	P@5	S@10	R@100	R@500	DM-AUC
BitcoinOTC	Random	–	–	0.495 [0.465, 0.525]	0.002	0.00	0.02	0.006	0.075	–
	Degree	none	0	0.910 [0.889, 0.930]	0.111	0.13	0.68	0.447	0.751	0.50
	Topology	none	0, 7–9	0.908 [0.886, 0.928]	0.116	0.11	0.67	0.456	0.741	0.52
	Full-R	none	0–2, 7–10	0.901 [0.877, 0.922]	0.089	0.09	0.58	0.424	0.734	0.51
	ChronoTrace	hub (from release)	0–10	0.899 [0.875, 0.921]	0.073	0.08	0.52	0.406	0.730	0.55
	ChronoTrace	ID cohort (from release)	0–10	0.928 [0.911, 0.944]	0.112	0.11	0.62	0.516	0.777	0.70
	Temporal	exact	3, 4, 10	0.939 [0.927, 0.951]	0.101	0.13	0.63	0.490	0.772	0.78
	Sgn-Struct	exact	1, 2, 5–9	0.946 [0.933, 0.957]	0.145	0.16	0.73	0.536	0.825	0.80
	ChronoTrace	exact	0–10	**0.949** [0.937, 0.960]	0.145	0.19	0.72	0.575	0.826	0.81
BitcoinAlpha	Random	–	–	0.474 [0.436, 0.512]	0.004	0.00	0.03	0.016	0.119	–
	Degree	none	0	0.869 [0.838, 0.898]	0.087	0.08	0.50	0.369	0.718	0.50
	Topology	none	0, 7–9	0.871 [0.842, 0.898]	0.071	0.07	0.48	0.358	0.715	0.53
	Full-R	none	0–2, 7–10	0.868 [0.838, 0.895]	0.067	0.07	0.40	0.347	0.711	0.52
	ChronoTrace	hub (from release)	0–10	0.856 [0.822, 0.888]	0.061	0.05	0.35	0.334	0.702	0.61
	ChronoTrace	ID cohort (from release)	0–10	0.880 [0.847, 0.909]	0.070	0.07	0.45	0.374	0.726	0.57
	Temporal	exact	3, 4, 10	0.907 [0.885, 0.928]	0.092	0.09	0.53	0.419	0.801	0.80
	Sgn-Struct	exact	1, 2, 5–9	0.930 [0.912, 0.947]	0.127	0.11	0.60	0.534	0.843	0.81
	ChronoTrace	exact	0–10	**0.931** [0.914, 0.947]	0.103	0.11	0.58	0.554	0.837	0.81
Epinions	Random	–	–	0.479 [0.451, 0.508]	0.000	0.00	0.00	0.000	0.004	–
	Degree	none	0	0.924 [0.903, 0.944]	0.051	0.10	0.35	0.189	0.401	0.50
	Topology	none	0, 7–9	0.926 [0.904, 0.946]	0.034	0.05	0.23	0.173	0.409	0.55
	Full-R	none	0–2, 7–10	0.922 [0.900, 0.942]	0.034	0.05	0.27	0.157	0.413	0.46
	ChronoTrace	hub (from release)	0–10	0.919 [0.898, 0.939]	0.029	0.04	0.27	0.164	0.390	0.48
	ChronoTrace	ID cohort (from release)	0–10	0.926 [0.905, 0.945]	0.028	0.04	0.22	0.155	0.399	0.59
	Temporal	exact	3, 4, 10	0.903 [0.879, 0.927]	0.022	0.04	0.27	0.128	0.342	0.61
	Sgn-Struct	exact	1, 2, 5–9	**0.926** [0.905, 0.946]	0.051	0.08	0.35	0.209	0.466	0.60
	ChronoTrace	exact	0–10	0.923 [0.899, 0.944]	0.030	0.04	0.22	0.173	0.428	0.59

differences behind these comparisons; for both window-free sets they scatter around zero.

~~The explanation is that the Temporal ablation uses only features 3, 4,~~

The ranking is useful for narrowing a search but rarely pinpoints a neighbor. With degree alone, the share of former neighbors among the top five candidates is 0.13, 0.08 and 0.10, and ~~10 (window activity ratio, pre/post shift, temporal concentration) and does not include any structural features such as degree , neighbor degree, two hop reach, or clustering proxy. On Epinions, where structural features carry more signal than temporal features (as evidenced by the high degree proportional null AUC of 0.912~~a former neighbor appears in the top ten for 68%, 50% and 35% of targets. Appendix D.1 reports precision, success and ranks at further depths.

A directory release adds an exact channel. In such a release a user left without events must have interacted only with the erased user. Such isolated users make up 2.3%, 3.9% and 6.6% of all former neighbors, and at least one appears for 15%, 25% and 48% of targets. Ranking by degree places them last, so a directory release also lowers the Degree adversary's AUC to 0.895, 0.838 and 0.830. Appendix E reports every adversary on directory releases.

Answer. The published graph lets an adversary rank former neighbors far above chance, with low top-of-list precision. Keeping isolated accounts visible discloses some former neighbors exactly. In practice, erasure does not hide a user's partners from someone who knows the user existed: on BitcoinOTC the 500 best-ranked of 5,880 candidates contain 75% of former neighbors.

6.2 RQ2: What explains inference without a window?

Degree explains the window-free adversaries. Even with every window-free feature (Full-R), ~~excluding structural features causes the Temporal group to underperform even the simple Degree baseline. This finding reinforces the paper's central claim that structural prominence dominates on large dense graphs. We caution that all~~

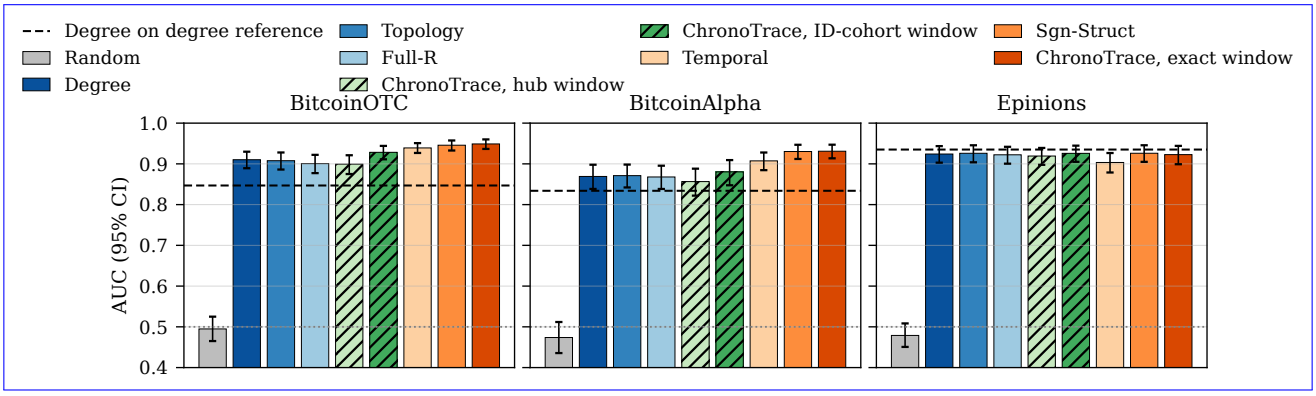


Figure 3: AUC of every adversary (95% CI over 60 targets). Blue: no window; green: window estimated from the published graph; orange: exact window. Dashed: Degree on the degree reference; dotted: chance.

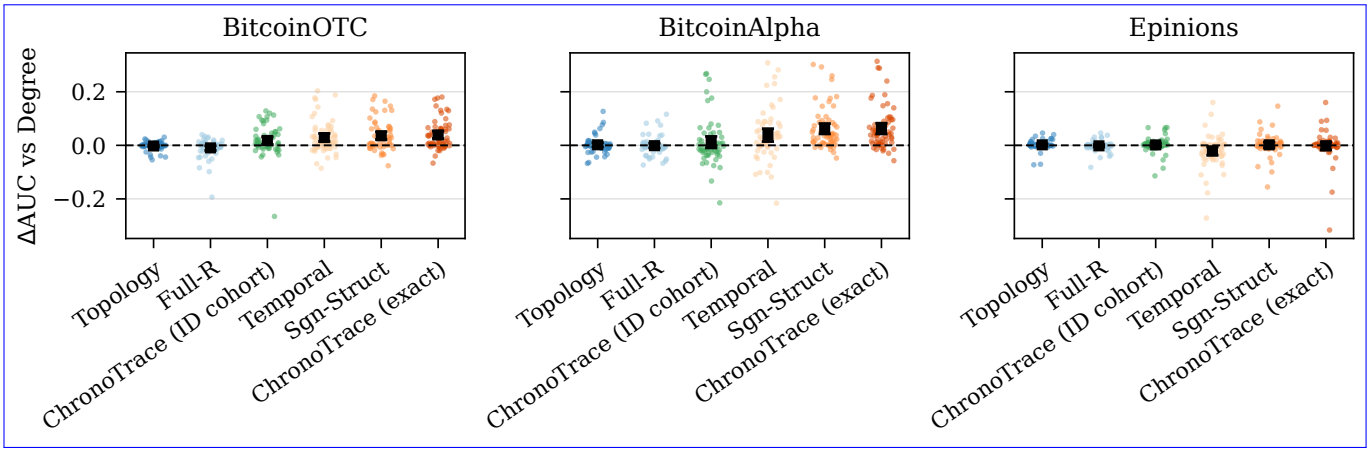


Figure 4: Per-target AUC difference to the Degree adversary (points), with mean and 95% CI (black).

Table 5: AUC on the real graph and on reference graphs (rewired: mean over independent rewirings), with degree controls; 60 targets per dataset. Degree (labels): label reference. DM-AUC: degree-matched AUC.

			Reference-graph AUC				
Dataset	Feature set	Real AUC	Degree (labels)	Degree (rewired)	Activity (rewired)	DM-AUC [95% CI]	Excess over degree
BitcoinOTC	Degree	0.910	0.856	0.847	0.844	0.50 [0.50, 0.50]	0.000
	Full-R	0.901	–	0.847	0.849	0.51 [0.45, 0.59]	−0.001
	ChronoTrace (exact)	0.949	–	0.843	0.932	0.81 [0.76, 0.86]	+0.079*
BitcoinAlpha	Degree	0.869	0.852	0.834	0.829	0.50 [0.50, 0.50]	0.000
	Full-R	0.868	–	0.837	0.828	0.52 [0.46, 0.58]	+0.006
	ChronoTrace (exact)	0.931	–	0.836	0.914	0.81 [0.77, 0.85]	+0.109*
Epinions	Degree	0.924	0.913	0.935	0.943	0.50 [0.50, 0.50]	0.000
	Full-R	0.922	–	0.935	0.940	0.46 [0.39, 0.52]	+0.002
	ChronoTrace (exact)	0.923	–	0.932	0.948	0.59 [0.52, 0.65]	+0.011

* $p < 0.05$, two-sided sign-flip permutation test (unadjusted).

~~Epinions temporal and oracle-window values are computed over~~ the ~~proxy record-order time axis (Section ??)and therefore reflect record-order locality rather than genuine interaction timing; the corresponding BitcoinOTC and BitcoinAlpha values are the ones that support timing-based conclusions.~~ degree-matched AUC is 0.51, 0.52 and 0.46, and every confidence interval contains 0.5 (Table 5). Full-R's excess over degree is −0.001, +0.006 and +0.002, none significant. Topology's degree-matched AUC is 0.52 [0.45, 0.58], 0.53 [0.47, 0.59] and 0.55 [0.49, 0.61], so its intervals contain 0.5 as well. Window-free features thus add nothing to degree.

Table 6: Mean standardized logistic coefficients over 60 targets (OTC: BitcoinOTC; Alpha: BitcoinAlpha; Epin.: Epinions).

		Full-R			ChronoTrace (exact)		
#	Feature	OTC	Alpha	Epin.	OTC	Alpha	Epin.
0	Degree	+1.33	+0.89	+1.01	+1.21	+0.49	+0.93
1	Sign balance	−0.11	+0.18	+0.14	+0.11	+0.50	+0.12
2	Direction balance	+0.30	+0.14	+0.05	+0.27	+0.13	+0.06
3	Window activity share				+1.31	+0.71	+0.37
4	Activity shift at window				+0.01	+0.17	+0.04
5	Positive window share				+0.46	+1.07	+0.52
6	Negative window share				+0.14	+0.24	+0.13
7	Neighbor degree	−0.39	−0.87	−1.23	−0.31	−0.98	−1.03
8	Two-hop reach	+0.22	+0.86	+1.70	+0.10	+1.02	+1.43
9	Closure ratio	−0.05	−0.06	−0.12	−0.06	−0.05	−0.08
10	Activity concentration	−0.14	−0.01	−0.15	−0.59	−0.38	−0.29

~~AP values are low (0.070–0.157 for chronotrace), reflecting extreme class imbalance (e.g., approximately 4–5 true neighbors among ∼5, 880 candidates on BitcoinOTC, yielding a positive-to-negative ratio of approximately 1:1,200). AUC is~~ The degree reference shows how much of the Degree adversary's advantage over chance any graph with the same degrees would give (Figure 5). On the rewired graphs, ranking by degree reaches 0.847, 0.834 and 0.935, against 0.910, 0.869 and 0.924 on the real graphs, so most of the advantage over chance is present in any graph with the same degrees. On the ~~primary metric; AP and Recall@k can behave nonmonotonically under this imbalance. On Epinions, degree alone achieves the highest AP (0.157), marginally outperforming chronotrace (0.156), further confirming that structural prominence dominates on this dataset. Operationally, Recall@500 ≈ 0.8 means the top-500 candidates capture roughly 4 of 5 true deleted neighbors.~~

~~Figure 3 visualizes the same results with bootstrap 95% CIs and null model calibration lines. The central observation is that nonoracle structural features (degree, topology, full nonoracle) already support deleted neighbor inference far above random, with the full nonoracle set representing the best result among the evaluated postrelease only supervised feature sets; oracle window temporal features produce a further but measured increment above the best evaluated nonoracle result~~Bitcoin graphs the gaps that remain, +0.063 (95% CI [+0.042, +0.083]) and +0.035 (95% CI [+0.001, +0.068]), mean that the erased users' partners are somewhat better connected than random partners of users with the same degrees; on Epinions the gap of −0.011 (95% CI [−0.032, +0.008]) is not significant. As Section 4.4 explains, these gaps describe how partners were chosen in the original graph rather than anything the deletion creates. On directory releases the same gap can turn negative, because ranking by degree places the isolated former neighbors of Section 6.1 last (Appendix E).

6.3 ~~Null model calibration~~

~~A natural concern is whether the high AUC of post-deletion degree is a genuine privacy leak or a tautological consequence of heavy-tailed degree distributions: in any network with~~

Table 7: AUC of former neighbors grouped by their degree in G_u^{rel}, each group against all non-neighbors.

	BitcoinOTC			BitcoinAlpha			Epinions		
Degree in G_u^{rel}	1–3	4–20	>20	1–3	4–20	>20	1–3	4–20	>20
Degree	0.30	0.76	0.98	0.28	0.74	0.96	0.49	0.89	0.99
Full-R	0.39	0.74	0.97	0.32	0.73	0.96	0.49	0.88	0.99
ChronoTrace (exact)	0.69	0.87	0.98	0.71	0.84	0.97	0.50	0.88	0.99
Former neighbors	23	97	423	26	108	309	64	137	767

~~preferential attachment, high-degree nodes are more likely to be neighbors of any given node. To calibrate this, we evaluate a null model in which "deleted neighbors" are selected proportional to degree in the *released* graph rather than by actual adjacency. This ensures the null model uses the same degree distribution visible to the attacker. If the degree-ranking AUC under the null model matches the observed AUC, the signal is entirely explained by degree structure rather than by actual deletion residues.~~ The model coefficients point the same way (Table 6). In the Full-R model the largest weights fall on degree or on two-hop reach, which grows with degree. Neighbor degree carries a negative weight in every model, so the model prefers well-connected users whose own neighbors are not hubs. Because the features are correlated, the weights show the direction of each feature's influence rather than its unique contribution, and the AUC confirms that together they add nothing to degree.

~~Table ?? reports the calibration. In each of 200 Monte Carlo trials per task, $|N_u|$ candidates are drawn without replacement from the released-graph candidate pool with probability proportional to released-graph degree, and the degree ranker is scored against them. Under a uniform random null,~~

Ranking by degree has a blind spot: it misses low-degree former neighbors (Table 7; Appendix D.7). It places former neighbors with one to three events at AUC 0.30, 0.28 and 0.49, at or below chance, and those with more than 20 events at 0.96 or above. Figure 1(c) illustrates the risk: a hub such as d, never linked to u, can outrank true former neighbors. Without a window, the attack therefore exposes mainly a user's relationships with well-connected users.

Answer. On the three dated graphs, an adversary without a window learns nothing detectable about the erased user beyond the degree of each candidate, and the degree ~~achieves AUC near 0.500 on all three datasets, confirming that degree carries no information when adjacency is independent of degree. Under the degree-proportional null, it achieves 0.855, 0.846, and 0.912, which~~ reference reproduces most of its advantage over chance. The risk falls on relationships with well-connected users.

6.3 RQ3: How much does knowing when help, and how accurately?

Exact timing adds signal that degree cannot explain. With the exact window, ChronoTrace reaches AUC 0.949 on

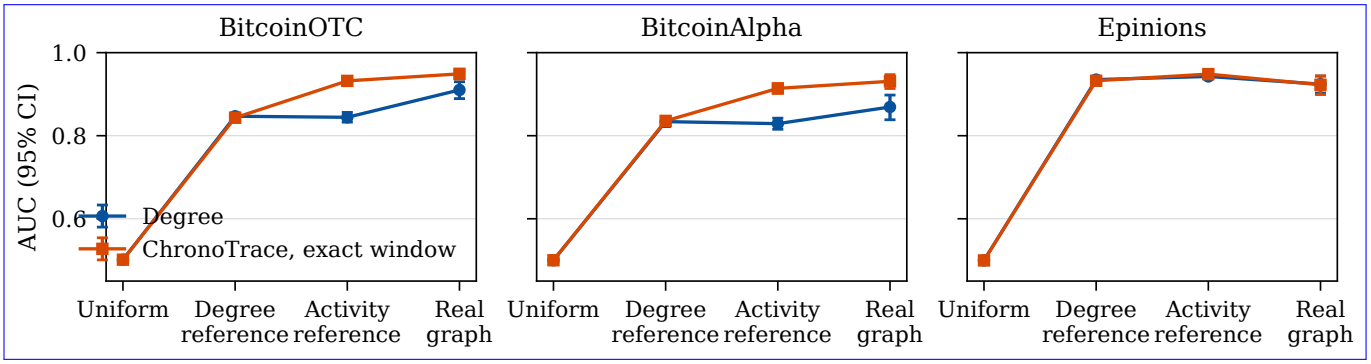


Figure 5: ~~Supervised ablation~~ AUC ~~with bootstrap 95% CIs of the~~ Degree and ~~null model calibration lines across three datasets. Nonoracle ablations~~ ChronoTrace (~~blue~~exact window) ~~carry strong signal above both null models. The degree-proportional null (orange dashed) shows that most degree signal reflects heavy-tailed structure, while~~ adversaries when the ~~gap between the null~~ former neighbors are drawn uniformly at random, drawn at random given degrees or given every user's activity, and ~~actual degree AUC measures leakage beyond that structure; this gap is positive only on BitcoinOTC. Mixed ablations~~ real (~~green~~95% CI)~~add oracle window temporal gain.On Epinions, the degree-proportional null (0.912) nearly matches chronotrace (0.928).~~

BitcoinOTC and 0.931 on BitcoinAlpha, gains of +0.039 and +0.062 over Degree (Table 4). Its degree-matched AUC is 0.81 and 0.81, and its excess over degree is +0.079 and +0.109 (both $p < 0.001$). It also ranks low-degree former neighbors, which degree misses, above chance, at AUC 0.69 (95% CI [0.63, 0.76]) and 0.71 (95% CI [0.64, 0.78]). The window features carry large weights: the share of a candidate's events inside the window is the largest coefficient on BitcoinOTC (+1.31), and the share of its positive events inside the window is the largest on BitcoinAlpha (+1.07).

The reference graphs show where this signal comes from (Figure 5, Table 5). On the degree reference, ChronoTrace reaches only 0.843 and 0.836. On the activity reference, which keeps every user's per-bin activity but randomizes partners, it reaches 0.932 and 0.914. The activity reference thus reproduces 84% and 82% of ChronoTrace's advantage over the degree reference: most of the ~~observed signal~~ timing signal comes from users being active in the same months. The gap to the activity reference, +0.017 (95% CI [+0.004, +0.030]) and +0.017 (95% CI [+0.001, +0.031]), is accuracy that per-bin activity does not explain. Like the gap to the degree reference, it can reflect reciprocity, community structure or target selection as well as the erased user's own relationships. The trading graphs record shared activity as reciprocal ratings: 68% of BitcoinOTC ratings are reciprocated within a week. By contrast, on Epinions the window does not improve the ranking (0.923 against 0.924 for Degree), its signal among candidates of equal degree is weak (degree-matched AUC 0.59), and both reference graphs score at least as high as the real one. Epinions trust statements are reciprocated far less often (Table 3).

~~The gap between the actual and the degree-proportional null is therefore the quantity of interest, and it does not have the same sign everywhere. On BitcoinOTC the gap is +0.041: deleted neighbors are detectably *more* structurally prominent than degree-proportional selection predicts, which is genuine deletion leakage beyond structure . On BitcoinAlpha it is −0.008 and on Epinions −0.044, so degree-proportional random selection actually *outperforms* degree ranking of the true deleted neighbors, and no leakage beyond the structural prior is detectable. Figure ?? visualizes the same comparison.~~

The gain requires a window that covers the user's active months (Figure 6; Appendix D.4). A window shifted by one bin, about two months, keeps the full gain, because with the one-bin padding of W_u it still covers every bin in which u was active. A shift of two bins, which misses part of that period, removes all of the gain on BitcoinOTC (+0.001 over Degree, not significant) and two thirds of it on BitcoinAlpha (+0.021, 95% CI [+0.009, +0.035]). A shift of eight bins removes it on both and leaves the BitcoinOTC adversary below Degree (−0.017).

6.4 ~~Adapted heuristic baselines~~

~~The supervised ablations above quantify how much signal is captured by a learned combination of engineered features. To understand whether simpler, unsupervised approaches can achieve comparable performance, we now evaluate adapted classical link prediction heuristics. Table 9 reports these adapted heuristic baselines for BitcoinOTC and BitcoinAlpha. Epinions is omitted from this table because computing anchor-set heuristics (which require dense $O(|V|)$ intersection operations per candidate) over the full 131k-node universe is computationally prohibitive under the 5-seed, 60-task protocol. The only nonoracle heuristic is degree prior, which achieves AUC 0.893 ± 0.102 on BitcoinOTC and 0.834 ± 0.150 on BitcoinAlpha . All other heuristics use oracle window anchor sets or burst/shift terms.~~ A release-only adversary can instead estimate the window

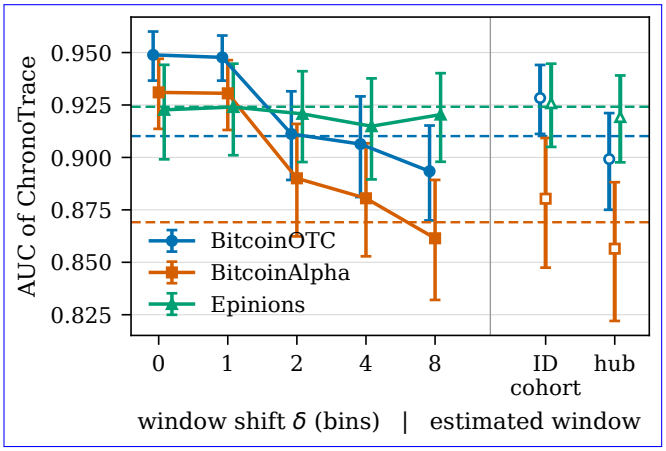


Figure 6: ~~Null model calibration across three datasets. Left bar: degree~~ **AUC** ~~when neighbors are selected uniformly at random~~ of ChronoTrace as its window is shifted by δ bins (~~near chance~~left) ~~. Center: degree AUC when neighbors are selected proportional to degree. Right: actual observed degree AUC. The gap between~~ or estimated from the ~~degree-proportional null and the observed AUC represents genuine deleted neighbor leakage beyond the baseline degree structure. On Epinions, the null~~ published graph (~~0.912~~right)~~exceeds actual degree AUC (0.868)~~. Dashed: Degree adversary.

from the published graph. The hub estimator, which gives every target the same window (IoU 0.26 on BitcoinOTC), does not beat Degree. By contrast, the ID-cohort estimator succeeds where IDs follow the order of first activity. On BitcoinOTC, where ID and first activity have Spearman correlation 0.998, its windows reach IoU 0.66. They raise AUC over Degree by +0.018 (95% CI [+0.003, +0.031]), about half of the exact-window gain, using only the published graph and the erased user's pseudonym. On BitcoinAlpha, where the correlation is 0.334, the ID-cohort window still carries signal beyond degree (excess over degree +0.040, $p < 0.001$) but does not significantly raise AUC over Degree (+0.011). On dated Epinions, whose IDs we assigned in order of first rating, its excess over degree is small (+0.011). ID order is thus the one route beyond degree that we found for a release-only adversary.

~~Preferential attachment and degree prior produce identical rankings because the anchor set mean degree $\bar{d}_A$ is constant per task, making preferential attachment a monotonic rescaling of degree (see Appendix ?? for proof). ChronoTrace unsupervised (CT-Unsupervised) achieves the highest heuristic AUC (0.915/0.872) but requires oracle window information.~~ **Answer.** Timing knowledge reaches relationships that degree cannot, including those of low-degree users, as long as the window covers the user's active months, and mainly on graphs whose edges record joint activity. Sequential pseudonyms supply enough timing on BitcoinOTC to recover about half of the gain without any side information, so whoever numbers the users of a release, platform or custodian, should not number them in order of first activity.

6.4 RQ4: Who is exposed, and what else is revealed?

~~Adapted heuristic baselines. Degree prior (nonoracle) is competitive with oracle window heuristics. Temporal burst and shift alone (oracle) are weak, confirming that simple raw temporal features without structural context are insufficient. Preferential attachment coincides with degree prior because the anchor set degree term is constant per task.~~

6.5 ~~GNN representation baseline: transfer failure~~

~~A natural question is whether off-the-shelf learned graphneural network representations, optimized for a related but different task (trust sign prediction), transfer to deleted neighbor inference. To test this, we evaluate DGTEN , a temporal trust-prediction GNN, as a representation baseline across the full 60-task benchmark on both BitcoinOTCand BitcoinAlpha (5 seeds, 12 nodes per seed each; Table 21). On BitcoinOTC, DGTEN embeddings alone achieve AUC 0.826 ± 0.152, substantially below degree (0.896 ± 0.098; paired delta -0.069 95% CI: -0.104, -0.040) . On BitcoinAlpha, DGTEN alone achieves AUC 0.737 ± 0.177, well below degree~~ On the trading graphs, users with few relationships are ranked no less accurately by AUC (~~0.838 ± 0.145; paired delta -0.101 -0.147, -0.057). On both datasets, combining DGTEN with hand-engineered features degrades performance relative to the hand-engineered features alone: on BitcoinOTC, ChronoTrace+DGTEN achieves AUC 0.932 ± 0.068 versus 0.946 ± 0.050 for ChronoTrace alone (paired delta -0.014 -0.025, -0.005); on BitcoinAlpha, ChronoTrace +DGTEN achieves AUC 0.891 ± 0.079 versus 0.930 ± 0.063 for ChronoTrace alone (paired delta -0.039 -0.057, -0.024). Notably, DGTEN's AUC of 0.737 on BitcoinAlpha falls below the degree-proportional null model (0.846).~~

~~These results demonstrate that GNN embeddings optimized for trust sign prediction do not transfer to residual leakage detection. They do not show that GNN architectures are inherently unsuitable, and we set out that distinction and its consequences below~~Figure 7). When the 60 targets are split into three groups of about 20 by the number of their former neighbors that remain in the release, ChronoTrace reaches AUC 0.955 for the smallest neighborhoods on BitcoinOTC and 0.956 on BitcoinAlpha, no lower than for the largest (0.937 and 0.916). By contrast, on Epinions, where timing does not help, the smallest neighborhoods are the least exposed (0.898 against 0.951). For targets with only one to three events, ChronoTrace reaches 0.978 and 0.956 on the Bitcoin graphs, against 0.953 and 0.893 for Degree, although their former neighbors

reach the top ten less often than those of the main targets (Appendix D.5).

The sign of a deleted relationship is also inferable. For every former neighbor present in G_u^{rel}, we use its remaining sign balance (feature 1) to predict whether its relationship with u included a negative rating. The prediction reaches AUC 0.82, 0.82 and 0.86; distrust occurs in 14%, 5% and 23% of these relationships. Users with many negative ratings are likely to have exchanged one with u too, so an adversary who finds a former neighbor can often also tell whether the two distrusted each other.

~~*DGTEN limitations.* The consistent transfer failure is not evidence that GNNs are inherently unsuitable for deleted neighbor inference. DGTEN was designed for trust sign prediction; its learned representations capture trust dynamics rather than neighborhood-structure residues.~~ The time of a deleted relationship is inferable as well. For each identified former neighbor, we take the time bin with the most events in its remaining record and compare it with the bin of its earliest deleted event with u. The two coincide for 43% of former neighbors on BitcoinOTC, 47% on BitcoinAlpha and 31% on Epinions, against 9%, 26% and 4% when always guessing the dataset's busiest bin, and they lie within one bin of each other for 61%, 69% and 47%. ~~A purpose-built GNN for link prediction or neighbor inference might perform differently. We include DGTEN as the only available temporal trust GNN with compatible architecture, not as a representative benchmark of GNN capability for this task; a fair comparison would require training a GNN specifically for deleted neighbor inference, a task for which no established architecture or training protocol exists. Epinions was excluded because it is a static snapshot with only proxy record-order timestamps (Section ??), which does not align with DGTEN's dynamic temporal modeling assumptions, limiting the GNN comparison to cryptocurrency networks~~window-informed adversary that takes the neighbor's busiest bin inside W_u comes within one bin for 75%, 80% and 62% of them, against 56%, 50% and 31% for the middle bin of W_u (Appendix D.6).

6.5 ~~Defense stress test~~

~~The results so far establish that deleted neighbor inference is feasible from the released graph alone. We now ask whether a platform can materially reduce this leakage through simple temporal edge suppression. Table 8 reports as a diagnostic stress test. On BitcoinOTC, temporal scrubbing reduces fused AUC from 0.943 to 0.923; on BitcoinAlpha , no variant meaningfully reduces AUC. Epinionsis omitted from this table because computing the iterative neighborhood intersections required by over its large 131k-node universe is prohibitively expensive for a diagnostic baseline. is a simple platform-side suppression baseline; its failure reinforces that local temporal edge removal is insufficient.~~ **Answer.** On the trading graphs, AUC does not fall with the size of a user's neighborhood, and the published graph indicates the sign and the approximate time of a deleted relationship as well as its likely existence. Erasure requests from users with few relationships therefore deserve the same protection as those from active users.

~~*Toward stronger defenses: why uniform edge deletion fails.* does not provide formal privacy guarantees. To evaluate a defense with formal guarantees, we implemented an edge-level randomized response (RR) mechanism. Standard RR provides ϵ-differential privacy by retaining true edges with probability $p = e^{\epsilon}/(1+e^{\epsilon})$ and flipping non-edges to fake edges with probability $1-p$. Because adding dense fake edges from the $O(|V|^2)$ non-edge universe is computationally prohibitive and ruins graph sparsity, we evaluate a *deletion-only* variant that drops edges with probability $1-p$ but adds no fake edges. While this relaxes the strict ϵ-DP indistinguishability guarantee (as missing edges could be true negatives or dropped positives), it serves as a rigorous uniform subsampling baseline. We evaluated the degree-only attacker against this defense across $\epsilon \in \{0.5, 1.0, 2.0, 4.0\}$ on all three datasets (Chronotrace attacker results are omitted because if a defense cannot break the fundamental degree prior, it fails as a comprehensive defense).~~

6.5 RQ5: Do simple defenses help?

~~The results demonstrate that uniform edge deletion cannot break the structural prior. Even at a highly restrictive~~ Neither defense of Section 4.6 changes the picture (Table 8, Figure 8). ChronoScrubremoves up to 14% of the events. Because it deletes events of the former neighbors, it lowers the Degree adversary's AUC by up to 0.044, yet ChronoTrace loses at most 0.016. Randomized response at $\epsilon = 0.5$ ~~(where only 62% of edges are retained), the degree leak AUC remains nearly identical to the undefended baseline. On BitcoinOTC , AUC shifts from 0.895 (undefended)to 0.893. On BitcoinAlpha, AUC shifts from 0.838 to 0.836. On Epinions, AUC shifts from 0.868 to 0.869. The degree prior is highly robust to random uniform subsampling: heavy-tailed nodes remain heavy-tailed even when uniformly decimated. A defense sufficient to resist the attack must break the degree-signal correlation directly, which requires either degree-preserving edge perturbation or k-anonymity on degree sequences. The benchmark provides the threat model against which such defensesshould be evaluated~~removes 38% of all events yet leaves the Degree adversary at 0.904 on BitcoinOTC (0.910 without it), because random subsampling preserves which users are well connected. Utility barely changes under either defense; on BitcoinAlpha, sign prediction is weak to begin with. We expect that a defense would need to flatten the degree heterogeneity of the published graph, for example through degree-sequence anonymization or differentially private graph synthesis [34]; we did not evaluate such defenses.

~~Per task AUC deltas vs. degree alone (60 tasks per dataset). Each point is one deleted node task. Black bars show mean and bootstrap 95% confidence interval.~~

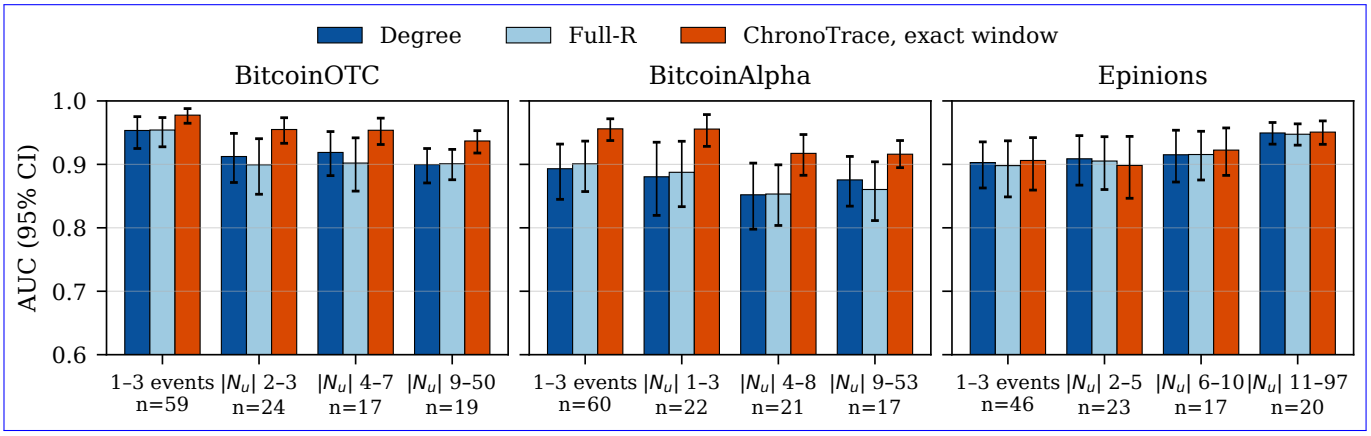


Figure 7: AUC by target group (95% CI): targets with one to three events, and the main targets split by their number of former neighbors $|N_u|$ that remain in the release.

Table 8: Defenses: mean AUC of the Degree and ChronoTrace (exact window) adversaries over 60 targets. Kept: share of events published. Utility: sign-prediction AUC on the defended release. RR: randomized response.

	BitcoinOTC				BitcoinAlpha				Epinions			
Defense	Degree	ChronoTrace	Kept	Utility	Degree	ChronoTrace	Kept	Utility	Degree	ChronoTrace	Kept	Utility
None	0.910	0.949	100%	0.635	0.869	0.931	100%	0.545	0.924	0.923	100%	0.767
ChronoScrub, temporal	0.875	0.933	91%	0.632	0.825	0.928	91%	0.544	0.920	0.918	86%	0.768
ChronoScrub, joint 0.25	0.901	0.943	97%	0.636	0.860	0.929	97%	0.545	0.928	0.926	96%	0.766
ChronoScrub, joint 0.50	0.895	0.942	94%	0.632	0.846	0.932	94%	0.549	0.928	0.926	92%	0.766
ChronoScrub, joint 0.75	0.886	0.939	92%	0.633	0.836	0.934	92%	0.550	0.920	0.921	88%	0.767
RR, $\epsilon = 0.5$	0.904	0.944	62%	0.639	0.856	0.925	62%	0.544	0.919	0.915	62%	0.767
RR, $\epsilon = 1$	0.907	0.946	73%	0.637	0.861	0.926	73%	0.544	0.927	0.924	73%	0.768
RR, $\epsilon = 2$	0.910	0.948	88%	0.637	0.867	0.930	88%	0.547	0.924	0.925	88%	0.764
RR, $\epsilon = 4$	0.910	0.949	98%	0.639	0.869	0.930	98%	0.548	0.923	0.922	98%	0.769

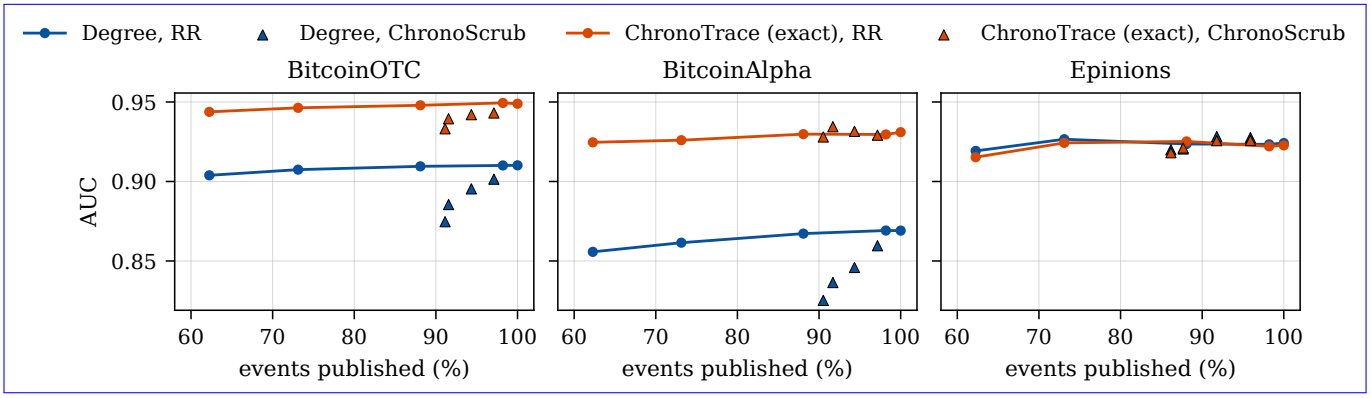


Figure 8: ~~stress test. Temporal local edge scrubbing changes edge retention and minimally affects sign prediction utility but does not eliminate reconstruction leakage~~ AUC against the share of events published, under randomized response (RR; $\epsilon = 4, 2, 1, 0.5$ from right to left) and the ~~oracle window attack~~four ChronoScrubvariants.

~~Positive values indicate improvement over degree alone. On BitcoinOTC, the only method whose CI excludes zero is chronotrace (the full oracle window model). On other datasets, several nonoracle methods also show significant gains over degree.~~ **Answer.** Local suppression and edge-level randomized response leave the degree signal and the timing signal largely intact. A platform therefore cannot substantially reduce the risk by deleting more edges around

the erased user; protection at the level of the published graph would likely require perturbing degrees themselves.

6.6 ~~Stratification by deleted degree~~

6.6 Adapted heuristic baselines

~~The aggregate results reported above average over 60 deleted-node tasks per dataset, but the risk may vary substantially depending on the deleted node's structural properties. Figure ?? stratifies AUC by the deleted node's incident degree. Tasks are divided into low degree (4 to 6), medium degree (7 to 12, or 14 depending on dataset), and high degree (13 or more, or 15 or more) strata, with the 60 tasks of each dataset distributed across the three strata. The oracle window chronotrace attacker achieves the highest AUC in every stratum on both datasets, and the nonoracle supervised ablations (degree, topology, full nonoracle)remain clustered together. The gap between chronotrace and the nonoracle ablations is largest in the high degree stratum, consistent with the interpretation that oracle window temporal features are most informative when the deleted node had more interactions to leave temporal residue. The per task scatter plot shows that AUC generally increases with deleted degree for all methods, though with substantial variance.~~ The adapted heuristics of Section 4.5 confirm these findings without training (Table 9; Appendix D.8). The neighbor-count prior, the only window-free heuristic besides the absolute sign balance, ranks close to the Degree adversary, and preferential attachment ranks identically to it. The anchor-based heuristics, which read the exact window and $|N_u|$, do not beat the trained ChronoTrace adversary; CT-Unsupervised comes closest on the Bitcoin graphs. The absolute sign balance ranks former neighbors below chance, mostly because users with one or two ratings have the largest absolute balance and are rarely former neighbors.

~~Deleted degree stratification (60 tasks per dataset). Left: AUC by degree stratum (low/medium/high) for selected ablations. Right: per task AUC vs. deleted degree (log scale). The oracle window chronotrace model achieves the highest AUC in every stratum. The nonoracle ablations are tightly clustered and do not separate by degree stratum.~~

7 ~~Discussion~~

~~The main implication is that record deletion and relational privacy are separable concerns. A platform may remove every edge incident to a deleted node yet release a graph in which the deleted node's neighbor set remains empirically inferable.~~

~~*Degree as a structural baseline.* The degree-proportional null model (Figure ??)is the key interpretive tool: it reveals that most degree signal reflects heavy-tailed structure rather than deletion residues. The genuine privacy finding is not that degree achieves high AUC, but that the structural prior *itself* encodes neighbor likelihood. Platforms cannot assume that deletion removes neighborhood information when degree distributions are heavy-tailed, even without any deletion-specific signal. The per-dataset gaps behind this reading are given in Section ??~~

6.1 Validity checks

Four checks address how the adversary and the evaluation were set up. First, training on erasures simulated on the original graph, whose training graphs still contain u, gives the same AUC as training on the published graph to within 0.0008 for every feature set on every dataset, in both release formats (Appendix C.2). Second, every result above uses the full candidate universe. On Epinions, AUC is unchanged when candidates are subsampled, but AP, P@k and the best rank degrade as the pool grows: P@5 of the Degree adversary falls from 0.34 with 2,000 candidates to 0.10 with all 56,498, so results reported on smaller pools overstate operational precision (Appendix E). Third, a random-forest attacker ranks no better (Appendix D.9), and the sign-binarization threshold does not change the results (Appendix D.10). Fourth, node embeddings from a variant of DGTEN, a temporal trust GNN, rank former neighbors worse than degree even though they are learned before erasure (Appendix D.11).

~~This dataset-dependent behavior creates a tension in our overall findings: while the dominant narrative is that the degree signal is primarily a structural artifact,~~

7 Discussion

What platforms and custodians can do. Erasure leaves the partners' popularity in place, and popularity already points to likely partners: on all three graphs, degree accounts for the accuracy of every window-free adversary (Section 6.2). Deleting more edges around the erased user does not help (Section 6.5), but a platform can measure the risk by running ChronoTrace's calibration on its own graph. If the degree reference reproduces most of the adversary's advantage over chance, as on all three of our graphs, the ~~BitcoinOTC result proves that genuine, detectable deletion leakage exists on some graphs. The direction of this gap depends on domain-specific topology. On large, dense graphslike Epinions, the structural prior overwhelms deletion residues; on sparser trust networks like BitcoinOTC, the residue of deletion remains detectable beyond the baseline structure. Both findingsunderscore that platforms must test their specific datasets rather than assuming either pure safety or pure vulnerability~~residual risk is the popularity of the user's partners, which we expect only degree-level perturbation of the published graph to reduce. Two cheaper measures follow directly from our findings. A platform that keeps accounts visible should hide users left without events after an erasure, because they disclose former neighbors exactly. And a platform or custodian should not number the users of a release in order of first activity, because sequential IDs supply timing knowledge that adds signal beyond degree.

~~*Operational attack utility*~~*When the findings transfer.* ~~AP values are low (0.070–0.157 for chronotrace)and P@5~~

Table 9: Adapted link-prediction heuristics (mean±std over 60 targets). R: release-only; W: reads the exact window and the number of former neighbors.

		BitcoinOTC			BitcoinAlpha			Epinions		
Heuristic		AUC	AP	R@500	AUC	AP	R@500	AUC	AP	R@500
Neighbor-count prior	R	0.907±0.086	0.104	0.742	0.865±0.126	0.084	0.714	0.922±0.083	0.052	0.391
|Sign balance|	R	0.283±0.095	0.001	0.024	0.280±0.133	0.001	0.011	0.208±0.103	0.000	0.007
Preferential attachment	W	0.907±0.086	0.104	0.742	0.865±0.126	0.084	0.714	0.922±0.083	0.052	0.391
Common neighbors	W	0.873±0.115	0.079	0.704	0.812±0.179	0.057	0.641	0.860±0.118	0.028	0.383
Jaccard	W	0.755±0.117	0.020	0.288	0.740±0.171	0.032	0.386	0.815±0.109	0.003	0.041
Adamic–Adar	W	0.879±0.115	0.099	0.731	0.821±0.178	0.071	0.660	0.862±0.118	0.033	0.393
Window activity share	W	0.773±0.130	0.004	0.156	0.747±0.152	0.006	0.283	0.580±0.187	0.000	0.002
Activity shift at window	W	0.758±0.146	0.041	0.513	0.699±0.176	0.042	0.465	0.775±0.153	0.013	0.240
Signed motif	W	0.922±0.070	0.073	0.757	0.898±0.090	0.060	0.728	0.880±0.115	0.032	0.358
CT-Unsupervised	W	0.929±0.061	0.057	0.774	0.905±0.085	0.058	0.776	0.917±0.087	0.017	0.295

ranges from 0.10 to 0.21 across datasets: an attacker inspecting the top-5 candidates correctly identifies a deleted neighbor in roughly 1 of 5 to 1 of 10 cases. This means that while the attacker can narrow down the search to a more promising subset (Recall@500 ≈ 0.8) , they do notgain a reliable single-target identification tool. For an investigating authority manually examining candidates, a false-positive rate exceeding 80% at top-5 would be operationally expensive. The practical value of the attack is therefore in narrowing down a candidate pool, not in pinpointing specific neighbors . An attacker seeking higher precision would likely need to combine this weak structural signal with auxiliary data Across our three graphs, the AUC of the Degree adversary orders with degree heterogeneity (Table 3): the more unequal the degrees, the more a degree ranking reveals. The window-informed results depend on whether an edge records joint activity, as the quickly reciprocated Bitcoin ratings do (Section 6.3) and unilateral endorsements, such as Epinions trust statements, do not. We expect other marketplaces with mutual feedback, such as peer-to-peer lending or ride sharing, to resemble the Bitcoin graphs, and follower or endorsement graphs to resemble Epinions.

Operational meaning. An adversary with the exact window narrows the candidates sharply but still needs further evidence: the share of former neighbors among its top five candidates is at most 0.19, and on BitcoinOTC the best-ranked former neighbor sits at median rank 4 (mean 20). For an investigator, following up a top-five list in which at least 81% of the candidates are not former neighbors would be expensive. Combining the ranking with other sources, such as cross-platform identity linkage, temporal correlation with a second platform or known events, or external metadata about the deleted node's interaction patterns. We do not evaluate such multi-source fusion in this work, but flag it as is a realistic escalation path that future defense workshould considerthat we leave to future work.

DGTEN comparison. Beyond the transfer failure itself (Section ??), the consistent degradation when DGTEN embeddings are combined with the hand-engineered featuressuggests that the embeddings introduce noise for this specific task rather than complementary signal.

Cross-domain scope. Evaluating on three datasets spanning two domains(cryptocurrency trust networks and consumer reviews) reveals that the degree-proportional null model calibration varies substantially across domains. On Epinions, the null (0.912) nearly matches the full chronotrace attacker (0.928), suggesting that on large, dense graphs, structural prominence alone explains most of the observable signal. This cross-domain variation underscores that single-dataset findings do not generalize even within the trust graph category.

Actionable recommendation for platform operators. Platforms relying on incident edge deletion as a privacy mechanism should audit whether their graph's degree distribution creates a structural prior that encodes neighbor likelihood. If the degree-proportional null model achieves high AUC on a representative sample of deleted nodes, then record deletion alone is insufficient for relational privacy and must be supplemented with structural perturbation (e.g., degree-preserving noise injection, edge-level randomized response, or differentially private graph synthesis). The benchmark provides a concrete auditing methodology for this purpose.

8 Limitations

Oracle window dependence. The strongest attack requires knowledge of the deleted interaction window; realistic attackers are limited to nonoracle ablations showing modest gains over degree alone. **Epinions proxy timestamps.** The public Epinions snapshothas no native timestamps; we assign proxy record-order timestamps (Section ??). Epinions temporal and oracle-window results therefore measure record-order locality, not interaction timing, and only its structural (degree , topology, null-model) results should be compared across datasets. Replacing Epinions with a genuinely timestamped signed graphis planned future work. **Small target sample.** 60 tasks per dataset limits formal significance testing; AUC differences below ∼0.05 should not be interpreted as robust. **Three-dataset; two-domain scope.** Generalization to other platforms, social networks, or collaboration graphs remains untested.

Low practical precision. AP ranges from 0.002 The adversary attacks one erasure in one published graph; several erasures, or two snapshots, would reveal more. The exact window is an upper bound, and only two estimators of it are evaluated. The adversaries are a logistic regression and a random forest on hand-built features, and the GNN embeddings we test were trained for sign prediction; an attacker that learns its own features for this task might find more. The three datasets come from two domains, all with binary signs; on the Bitcoin graphs, whose ratings range from -10 to 0.116 due to extreme class imbalance; the attack is a ranking measurement, not a high-precision identification tool. **Classifier sensitivity.** Gradient boosting and neural classifiers are not evaluated. Binary sign binarization sensitivity is tested in Appendix D.10 and found to have negligible effect. **GNN baseline scope.** DGTEN is the only evaluated GNN; a multi-architecture comparison is needed before concluding that hand-engineered features universally outperform learned representations. **Defense baseline.** is a diagnostic stress test, not a proposed defense. We additionally evaluate an edge-level randomized response (DP) baseline in Section ??, finding that edge deletion alone cannot break the structural prior. Stronger graph perturbation strategies remain unevaluated. $+10$, the binarization threshold does not change the results (Appendix D.10). Dated Epinions keeps only the statements with a genuine date, and our targets exclude users with more than 128 events. The reference graphs preserve degrees or activity but not reciprocity or community structure. On the static Epinions snapshot, structure ranks candidates of equal degree slightly above chance (Appendix E), so the finding that degree explains window-free inference may not hold for every graph. Each dataset contributes 60 targets drawn over degree quartiles, which bounds the precision of small differences; the confidence intervals report this. ChronoScrubis a diagnostic, the randomized response we evaluate only deletes events, and we did not evaluate defenses that perturb degrees. The secondary analyses of sign and time each use a single predictor.

9 Conclusion

This paper presented Erasing a user from a trust graph removes the user's ratings, while the counterparties' ratings still reveal part of the user's relationships. ChronoTrace , an empirical benchmark for deleted neighbor inference in temporal signed trust graphs. The central methodological contribution is degree-proportional null model calibration, which separates genuine deletion leakage from structural artifacts: platforms cannot assume that deleting a node's records removes neighborhood information when degree distributions are heavy-tailed. Oracle window temporal features provide the largest gain beyond this structural baseline, although on Epinions that gain is not statistically distinguishable from the best nonoracle feature set, and operational precision remains low throughout, meaning the attack narrows candidate pools rather than pinpoints specific neighbors. Future graph deletion defenses should therefore target the structural prior directly through degree preserving perturbation or degree sequence anonymization, rather than relying on edge level deletion or uniform subsampling, both of which leave the heavy tail signal intact. We position this work as an honest measurement benchmark that provides both the threat model and the calibration methodology against which future graph deletion defenses should be evaluatedmeasures this residual inference and attributes it. An adversary without timing knowledge ranks former neighbors well because they are well connected, and calibration shows that degree explains it. Knowing when the user was active reaches relationships beyond degree, including those with low-degree users, as long as the window covers the user's active months, and mainly on graphs that record joint activity; sequential pseudonyms can supply that timing from the published graph itself. Platforms should hide accounts left isolated by an erasure, avoid sequential pseudonyms, and use calibration to decide whether degree-level perturbation is needed.

Ethical Considerations

This work was conducted entirely on uses only public, previously released network datasets and involved involves no human subjects, so it was not submitted to an institutional review board. The BitcoinOTC and BitcoinAlpha signed trust graphs are distributed publicly through the SNAP repository [19], and the Epinions signed graph is a public static snapshot of the Epinions. com trust network. dated Epinions statements through the Network Repository [24]. All three contain only pseudonymous numeric node identifiers , with identifiers and no names, contact details , content, or other personally identifiable informationor content. We contacted no individualsone, collected no new dataabout any person, and report only aggregate metricscomputed over sampled deletion tasks; no per-node result identifies a specific account.

The benchmark measures an attack, so it ; no per-user result identifies an account. The method is dual-use, and we state that directly. An investigating authority could use candidate narrowing to identify former associates of a deleted node. An investigator or a stalker could use the ranking to find associates of an erased user, and a stalker could apply the same procedure to infer a deleted user's connections. Platform operators, conversely, can run the benchmark defensively to audit whether their own deletion pipeline leaves a node's neighborhood inferableplatform or custodian can use the same ranking to audit its erasure procedure. Three factors bound the incremental harmof publication. First, operational precision is consistently . Precision is low (P@5 of 0.10–0.21), so the method narrows a candidate pool rather than identifying a specific neighbor, and it offers an adversary no reliable single-target capability. Second, every component technique is already standard in public machine learning toolkits (degree ranking, classical link-prediction heuristics, and logistic regression), so we

disclose no new capability. Third, the strongest configuration requires oracle knowledge of the deleted node's interaction window, which a released-graph adversary does not have. Weighing these against the defensive value of a calibrated auditing methodology, and following the at most 0.19). Without timing knowledge, the adversary learns nothing beyond degree, which any public degree list already reveals. The two side channels we identify, isolated accounts and sequential pseudonyms, are cheap for platforms to close, and stating them helps platforms close them. Following the Menlo Report principles of beneficence and minimal risk [8], we judge that publication is publication to be net beneficial. We report the null-model calibration precisely because it prevents the structural prior from being overstated as an attack, which is the failure mode most likely to cause disproportionate alarm or misplaced confidence in deletion as a privacy mechanism.

Open Science

We support open science and commit to releasing the The complete ChronoTrace benchmark upon acceptance. We do not provide an artifact at submission time, so no repository is linked here; the release will instead be made at a stable, citable location under a permissive open-source license, and we will submit it to the PoPETs artifact evaluation process.

The release will comprise four parts. First, the benchmark implementation: is available at https://anonymous.4open.science/r/ChronoTrace-0265/. It contains feature extraction, the supervised attacker, the adapted heuristic baselines, the degree-proportional null-model calibration, and the and randomized-response defense stress tests. Second, the driver adversaries and their training on simulated erasures, the window estimators, the reference graphs and degree controls, the defenses, every per-target result file, and the scripts that produce every tableand figure , figure and number in this paper , together with the seeds and configuration used for the reported runs. Third, the result files, as per-task metric records in JSON, from which the reported means, standard deviations, and bootstrap confidence intervals are computed, so that each number in the paper can be traced to the task-level values behind it. Fourth, the preprocessing codefrom those files. The datasets are public and are not redistributed: the artifact documents how to obtain them, and a preparation script rebuilds the exact input files, including the construction of the Epinions proxy record-order time axis described in Section ??, so that its effect on the Epinions temporal results can be inspected directly.

We will not redistribute the datasets, which are already public: BitcoinOTC and BitcoinAlpha are used unmodified from the SNAP repository , and Epinions is the public static SNAP signed-Epinions snapshot. The release will document the download and preprocessing steps for eachdated Epinions statements, and verifies them against checksums. The artifact will be submitted to the PoPETs artifact evaluation.

AI Use

The authors used LLMs solely for proofreading, improving language and clarity, and organizing the LaTeX source. The research methodology, experiments, results, figures, and scientific content are designed the study, checked and verified the code and results, and are responsible for every statement in this paper. LLM-based assistants were used, under the authors' own contributions. The authors remain fully responsible for the accuracy, originality, and integrity of the manuscriptdirection, to organize the LaTeX source, draft and edit text, arrange the figures and tables, and implement and test analysis code for this revision.

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A ~~Extended Results~~

~~This appendix provides supplementary tables and figures that support the main body findings. Content here does not count toward the main body page limit.~~

A Feature Definitions

For candidate v in G_u^{rel}, let p_v and n_v be its numbers of positive and negative events, o_v and i_v its numbers of events given and received, and $h_v \in \mathbb{N}^{32}$ its events per time bin. (0) $\log(1+\deg v)$, where $\deg v$ counts v's events. (1) $(p_v - n_v)/\max(1, p_v + n_v)$. (2) $(o_v - i_v)/\max(1, o_v + i_v)$. (3) The share of v's events in $[b_0, b_1]$. (4) $|\bar{h}_v^{\mathrm{post}} - \bar{h}_v^{\mathrm{pre}}|$, the change in mean events per bin between the three bins after b_1 and the three bins before b_0, without normalization. (5, 6) The shares of v's positive and of its negative events in $[b_0, b_1]$. (7) $\log(1+\text{mean degree of } v\text{'s neighbors})$, zero without neighbors. (8) $\log(1+|T_v|)$, where T_v is the set of users exactly two hops from v. (9) $|N(v)|/\sqrt{1+|T_v|}$, a closure ratio that is large when v's neighborhood closes on itself (it is not the clustering coefficient). (10) $1 - H(h_v)/\log 32$, where H is the Shannon entropy of v's normalized bin counts. All features are standardized before fitting.

A.1 ~~Richer Ranking Metrics~~

B Experimental Protocol

This appendix specifies the protocol behind every result. The released code implements exactly these steps.

~~Table ?? reports Precision@k, Success@k, best positive rank, and mean positive rank for selected ablations. These metrics complement the AUC and AP results in Table 4 by characterizing ranked list quality at specific operating points.~~

B.1 Data

Each dataset is a list of rating events (rater, rated user, rating, time), sorted by time. A rating above zero is positive and any other rating is negative; Bitcoin ratings are never zero, and every Epinions statement is either trust or distrust. We use the events as released, without merging repeated pairs (none occur) or renumbering users, with one exception. The dated Epinions graph keeps every statement of the Extended Epinions release [22, 24] that carries a genuine creation date, without self-ratings, and numbers its users in order of first appearance: users who rated others come first, ordered by their first rating given, followed by users who

were only rated. BitcoinOTC and BitcoinAlpha keep the user IDs of their SNAP releases [19]. The static snapshot keeps its self-ratings and uses the record index as time. The 32 time bins divide the span from the first to the last event of each real graph into equal intervals, and the reference graphs use the same bins. Reciprocated ratings come back quickly on the Bitcoin graphs, where 86% and 84% of reciprocated pairs are reciprocated within a week, against 62% on Epinions.

~~Operationally, the chronotrace attacker achieves P@5 of 0.180 (OTC)~~

B.2 Targets, candidates and labels

We use the seeds 42, ~~0.100 (Alpha)~~ 7, ~~and 0.213 (Epinions)~~ 13, ~~so at best roughly one in five top-ranked candidates is a true deleted neighbor, and only one in ten on BitcoinAlpha. Its mean positive rank of 315 (OTC) , 267 (Alpha),~~ 100 and ~~369 (Epinions) is nonetheless substantially better than degree alone (619/611/677)~~ 200. For each seed, the main targets are three users drawn at random from each quartile of the incident-event count among the users with 4 to 128 events. The low-degree targets of Appendix D.5 are 12 users per seed drawn uniformly among the users with one to three events. A user drawn under two seeds counts as two targets; the main targets contain 1 such repeat. In an edge-list release, the candidates are the users with at least one event in the published graph, after any defense; a former neighbor left without events is neither a positive nor a negative and is counted as isolated. In a directory release, every user except u is a candidate. A target with no former neighbor among its candidates has no defined AUC and is left out; this happens only among low-degree targets (Appendix D.5).

B.3 Training the adversaries

For a target u drawn with a given seed, the shadow users are the other main targets of that seed: eleven for a main target and twelve for a low-degree target. Each shadow erasure deletes a shadow user u_j from G_u^{rel}. Its labels are u_j's neighbors in G_u^{rel}, and its exact window comes from u_j's events in G_u^{rel}, so no event, label or window of u enters training. The adversary pools the rows of all shadow erasures and keeps every positive. From the n_- negatives it draws $\min(n_-, \max(1000, 20n_+))$ without replacement, where n_+ counts the positives, using a generator seeded by the seed plus u's ID. It standardizes the features on the drawn rows and fits a logistic regression with an L2 penalty ($C = 1$), balanced class weights and at most 1,~~indicating that the full model improves the expected position of true neighbors. On Epinionsthe ordering reverses at small k: degree alone attains higher P@5 (0.230) and S@5 (0.617)than chronotrace, consistent with the degree-proportional null already explaining most of the observable signal on that dataset. Best positive rank of 18.6 (OTC), 19.5 (Alpha),~~ 000 iterations. The random forest of Appendix D.9 uses the same rows. The submitted protocol, compared in Appendix C.2, differs only in its training graphs: it deleted each shadow user from the original graph G, which still contains u.

B.4 Timing knowledge

The exact window spans the bins of u's first and last events in G, given or received, padded by one bin on each side and clipped to the 32 bins. A shift by δ moves the window by δ bins in a direction drawn at random for each target and keeps its width; near the ends of the time span the window moves only as far as the span allows. Each shadow window is shifted by the same δ in its own random direction. The hub estimator ranks the users of the published graph by their number of distinct neighbors, takes the median time of all published events that touch one of the top 50, and returns the seven bins centered on that time, truncated at the ends of the span. The ID-cohort estimator takes the 25 users of the published graph whose IDs are closest to u's and finds each one's first and last active bins. It returns the bins from the floor of the median first bin to the ceiling of the median last bin, padded by one bin. Both estimators treat the shadow erasures in the same way.

B.5 Metrics and statistics

AUC counts ties as one half, and ~~20.9 (Epinions)means the highest-ranked true neighbor typically appears within the top~~ AP is the non-interpolated average precision. Ranks start at 1 for the top candidate. The degree-matched AUC is the AUC over all pairs of a former neighbor and a non-neighbor that have the same number of events in G_u^{rel}. All intervals are percentile bootstrap intervals over targets with 10,000 resamples, and paired differences use the targets present in both arms. The p-values are two-sided and come from 10,000 random sign flips of the per-target differences. The Benjamini–Hochberg adjustment applies within each dataset to the eight comparisons of Table 13; all other p-values are unadjusted. The groups by $|N_u|$ in Figure 7 are the tertiles of the number of former neighbors present in the release. The pool-size analysis of Appendix E reuses the scores computed on the full universe, keeps every former neighbor, and averages ten random draws of non-neighbors for each pool size.

B.6 Defenses and utility

The residue of ChronoScrubconsists of N_u and every neighbor of N_u in G_u^{rel}. Each event of G_u^{rel} receives a drop score. At strength s, an event inside W_u that touches the residue scores $0.45s$, with $s = 1$ for the temporal variant. In the joint variant, an event between two residue users scores at least $0.25s$, at any time. A deterministic hash maps each event to a number in $[0, 1)$, and the event is deleted when that number is below its drop score. Randomized response keeps every event of G_u^{rel} independently with probability $e^{\epsilon}/(1 + e^{\epsilon})$, using a generator seeded by the seed plus u's ID.

Utility is measured by sign prediction. The test set consists of the latest 20~~candidates~~% of the events of the undefended release, of which at most 4,000 are sampled. The training set consists of the events of the defended release that precede them, of which at most 12,000 are sampled. Each event is described by eight features of its endpoints in the training graph, computed without the event itself: the logarithms of both endpoints' degrees, of the rater's out-degree and of the rated user's in-degree; both endpoints' sign balances; the logarithm of their number of common neighbors; and their Jaccard similarity. A logistic regression with balanced class weights predicts the sign, and utility is its AUC on the test set.

B.7 Sign and time of deleted relationships

Both analyses use every pair of a main target u and a former neighbor v that remains a candidate in the edge-list release. The relationship counts as distrust if any rating between u and v, in either direction, is negative. The sign predictor scores v by the negative of its sign balance in G_u^{rel} (feature 1), and the AUC pools all pairs. The time of the relationship is the bin of the earliest event between u and v. We compare three predictors of it: the dataset's busiest bin over all events of G; v's busiest bin in G_u^{rel}; and, for a window-informed adversary, v's busiest bin inside W_u. Ties go to the earliest bin, and intervals bootstrap over targets.

~~Per-Node Heterogeneity Analysis~~

B.8 Computation

All edge-list experiments together take about four hours of wall-clock time on up to 22 CPU worker processes. The GNN of Appendix D.11 trains in minutes on one GPU.

~~Figure ?? shows the per-node AUC distribution under the oracle window chronotrace attacker. Each point represents one of 60 deleted node tasks. AUC ranges from approximately 0.7 to 1.0 on both datasets, revealing substantial task-level heterogeneity driven by differences in deleted degree, local neighborhood density, and temporal span.~~

C Calibration Details

~~This heterogeneity has implications for privacy risk assessment: while the aggregate AUC of 0.946 (OTC) and 0.930 (Alpha) suggests consistently high inference, some deleted nodes are substantially more vulnerable than others. Nodes with fewer,more structurally distinct neighbors tend to have higher per-task AUC values.~~

C.1 Reference graphs

Both rewirings keep each event's rater, sign and time, and permute the rated users within groups: over all events for the degree reference, and among the events of the same time bin and sign for the activity reference. A permutation can map an event onto its own rater. We repair such self-loops by swapping rated users within the group and would drop any that remained; none did. Repeated rater–rated pairs are kept, so the rewiring preserves every user's event counts but neither its number of distinct neighbors nor the graph's reciprocity. We use five rewirings of each Bitcoin graph and three of Epinions, seeded identically for both references, and average each target's result over them. Table 10 reports the number of rewirings and the dropped and duplicated events. The label reference draws $|N_u|$ candidates without replacement with probability proportional to their degree in G_u^{rel}, 200 times per target, and scores the Degree ranking against them.

Table 10: Reference graphs: rewirings per dataset, and mean numbers of events dropped as self-loops and of repeated rater–rated pairs per rewiring.

Dataset	Reference	Rewirings	Self-loops	Duplicates
BitcoinOTC	degree	5	0	1539
BitcoinOTC	activity	5	0	2528
BitcoinAlpha	degree	5	0	1076
BitcoinAlpha	activity	5	0	1753
Epinions	degree	3	0	9965
Epinions	activity	3	0	13565

~~Per-node AUC under the oracle window chronotrace attacker (60 tasks per dataset, 5 seeds). Each point is one deleted node task. Substantial heterogeneity is visible, with AUC ranging from ~0.7 to 1.0.~~

C.2 ~~Precision-Recall Analysis~~

~~Figure ?? presents Precision~~

C.2 Protocol check

Table 11 compares the two ways of training the adversary. The submitted protocol trained on erasures of other targets from the original graph, so its training graphs still contained the evaluated target and its edges. The protocol used throughout this revision trains only on erasures simulated on the published graph. Every difference is below 0.001 AUC in both release formats, so the target's own information did not drive the results.

D Extended Results

This appendix gives every result behind the main text, organized by the question it supports.

D.1 Ranking metrics at fixed depths

Table 12 complements Table 4 with precision and success at the top of the ranking and with rank statistics, and Figure 9 plots P@k ~~and Recall~~k and R@k~~curves for selected ablations.~~ ~~The key observation is that the nonoracle ablations (degree, topology, full nonoracle) are tightly clustered on both datasets, consistent with the finding that nonoracle features beyond degree provide limited additional discriminative power. The chronotrace model separates clearly from the nonoracle cluster, with the gap being most pronounced at small k values where operational precision matters most .~~ ChronoTrace with the exact window leads or ties in precision and rank on the Bitcoin graphs, although Sgn-Struct has a

Table 11: AUC with training on erasures simulated on the published graph minus AUC with the submitted protocol, in both release formats (mean [95% CI] over 60 targets).

Release	Feature set	BitcoinOTC	BitcoinAlpha	Epinions
edge-list	Degree	0.0000 [0.0000, 0.0000]	0.0000 [0.0000, 0.0000]	0.0000 [0.0000, 0.0000]
	Topology	−0.0001 [−0.0006, +0.0002]	+0.0002 [−0.0003, +0.0007]	+0.0003 [+0.0001, +0.0006]
	Full-R	−0.0003 [−0.0010, +0.0003]	−0.0001 [−0.0007, +0.0005]	−0.0001 [−0.0006, +0.0004]
	Temporal	−0.0006 [−0.0016, 0.0000]	−0.0001 [−0.0003, +0.0002]	+0.0002 [−0.0001, +0.0006]
	Sgn-Struct	−0.0004 [−0.0017, +0.0004]	−0.0001 [−0.0009, +0.0005]	+0.0004 [−0.0001, +0.0011]
	ChronoTrace (exact)	−0.0001 [−0.0010, +0.0008]	−0.0001 [−0.0010, +0.0009]	+0.0004 [−0.0001, +0.0010]
directory	Degree	0.0000 [0.0000, 0.0000]	0.0000 [0.0000, 0.0000]	0.0000 [0.0000, 0.0000]
	Topology	−0.0003 [−0.0008, +0.0001]	+0.0002 [−0.0003, +0.0007]	−0.0004 [−0.0008, 0.0000]
	Full-R	−0.0004 [−0.0009, 0.0000]	−0.0001 [−0.0007, +0.0006]	+0.0001 [−0.0003, +0.0005]
	Temporal	−0.0001 [−0.0003, +0.0001]	−0.0008 [−0.0023, +0.0001]	+0.0001 [−0.0001, +0.0003]
	Sgn-Struct	−0.0004 [−0.0008, 0.0000]	+0.0001 [−0.0005, +0.0007]	−0.0001 [−0.0006, +0.0004]
	ChronoTrace (exact)	−0.0002 [−0.0009, +0.0005]	+0.0001 [−0.0008, +0.0010]	−0.0002 [−0.0008, +0.0004]

slightly higher S@10. On Epinions, Degree has the highest P@5 although the learned sets have similar AUC, because their gains come from the middle of the ranking rather than its top.

On BitcoinAlpha, the nonoracle ablations show slightly more separation than on BitcoinOTC, consistent with the finding that topology features provide a statistically detectable improvement on Alpha but not on OTC.

D.2 ~~Classifier Family Sensitivity~~

D.2 Paired comparisons

Table 13 lists every paired comparison against the Degree adversary with bootstrap intervals and Benjamini–Hochberg adjusted p-values. It is the numerical counterpart of Figure 4.

To assess whether the findings depend on the choice of logistic regression, we compare against random forest (100 estimators, max depth 10, balanced class weights) on the full 5-seed, 60-task evaluation. Table **??** reports the comparison.

Random forest consistently underperforms logistic regression, with the gap being largest for degree-only features (AUC drops from 0.896 to 0.753 on OTC)and narrowing with additional features (chronotrace: LR 0.943 vs. RF 0.919 on OTC). This is consistent with random forest requiring more per-leaf samples than extreme class imbalance provides. Critically, the qualitative conclusions are preserved across classifier families: degree dominates nonoracle signal, and chronotrace adds temporal gain.

D.3 Per-target heterogeneity

Figure 10 shows every target's AUC against its number of former neighbors. Exposure varies widely between users with few former neighbors and less between users with many.

D.4 ~~Bootstrap Confidence Interval Details~~

D.4 Timing knowledge

Table 14 reports ChronoTrace under every level of timing knowledge, with the overlap of each window with the exact one. Table 15 repeats the analysis for all three window-based feature sets. The pattern of Figure 6 holds for each of them: a one-bin shift changes nothing, a two-bin shift brings every set down to about the Degree adversary, and an eight-bin shift leaves Temporal, which has no degree feature, well below it (0.847 and 0.773 on the Bitcoin graphs).

Table **??** provides the full paired bootstrap 95% CI analysis for all pairwise method comparisons against the degree baseline. CIs are computed from 10, 000 resamples of per-task AUC values. A CI that excludes zero indicates a statistically distinguishable difference at the 95% level.

The pattern is clear: on BitcoinOTC, only oracle window methods produce statistically distinguishable improvements over degree. On BitcoinAlpha and Epinions, both nonoracle (topology, full nonoracle) and oracle window methods show statistically significant gains. The DGTEN embedding baseline shows the opposite pattern: it performs significantly worse than degree on both BitcoinOTC (paired delta −0.069) and BitcoinAlpha (paired delta −0.101), with both CIs excluding zero. This dataset-dependent result is a key finding: the degree baseline is harder to beat on BitcoinOTC, while on the other datasets, even modest feature additions provide measurable improvements.

D.5 Low-degree targets

Many users have only one to three events, so we also erase 60 such users per dataset, 12 per seed, drawn uniformly. The adversary trains on simulated erasures of the same seed's main targets, which supply more positive examples than low-degree users would. A target whose former neighbors all disappear from the edge-list release has no defined AUC and is left out of the averages; in a directory release those neighbors would be identified exactly. Table 16 reports the results.

D.6 ~~Null Model Calibration Details~~

The degree-proportional null model is constructed as follows: for each task, instead of using actual deleted neighborsas positives, we sample $|N_u|$ candidates from the released-graph candidate pool C_u without replacement, with sampling probability proportional to each candidate's degree in the released graph. This preserves the post-deletion degree distribution (

Table 12: Ranking metrics at fixed depths (mean over 60 targets). S@k: at least one former neighbor in the top k. Best and mean: rank of the best-ranked and the average rank of the former neighbors.

Dataset	Feature set	Window	P@5	P@10	P@20	S@5	S@10	Best	Mean
BitcoinOTC	Degree	none	0.13	0.11	0.08	0.48	0.68	35	533
	Topology	none	0.11	0.10	0.08	0.38	0.67	35	547
	Full-R	none	0.09	0.09	0.07	0.37	0.58	36	589
	ChronoTrace	hub (from release)	0.08	0.09	0.07	0.27	0.52	38	597
	ChronoTrace	ID cohort (from release)	0.11	0.10	0.08	0.42	0.62	23	426
	Temporal	exact	0.13	0.11	0.08	0.53	0.63	29	361
	Sgn-Struct	exact	0.16	0.13	0.10	0.58	0.73	33	323
	ChronoTrace	exact	0.19	0.14	0.10	0.58	0.72	20	305
BitcoinAlpha	Degree	none	0.08	0.07	0.06	0.37	0.50	67	494
	Topology	none	0.07	0.07	0.05	0.28	0.48	67	491
	Full-R	none	0.07	0.06	0.05	0.30	0.40	65	503
	ChronoTrace	hub (from release)	0.05	0.05	0.05	0.22	0.35	75	546
	ChronoTrace	ID cohort (from release)	0.07	0.06	0.05	0.25	0.45	73	456
	Temporal	exact	0.09	0.08	0.07	0.42	0.53	53	366
	Sgn-Struct	exact	0.11	0.09	0.08	0.43	0.60	30	267
	ChronoTrace	exact	0.11	0.09	0.08	0.43	0.58	30	264
Epinions	Degree	none	0.10	0.08	0.06	0.33	0.35	167	4246
	Topology	none	0.05	0.04	0.04	0.18	0.23	172	4195
	Full-R	none	0.05	0.04	0.04	0.18	0.27	182	4416
	ChronoTrace	hub (from release)	0.04	0.04	0.03	0.15	0.27	191	4572
	ChronoTrace	ID cohort (from release)	0.04	0.04	0.03	0.13	0.22	180	4194
	Temporal	exact	0.04	0.04	0.03	0.17	0.27	635	5608
	Sgn-Struct	exact	0.08	0.07	0.06	0.27	0.35	129	4189
	ChronoTrace	exact	0.04	0.04	0.04	0.18	0.22	160	4374

~~Precision and Recall at selected k values~~
~~(60 tasks per dataset, 5 seeds, mean ± SEM). The chronotrace model achieves the highest precision at small k on both datasets.~~

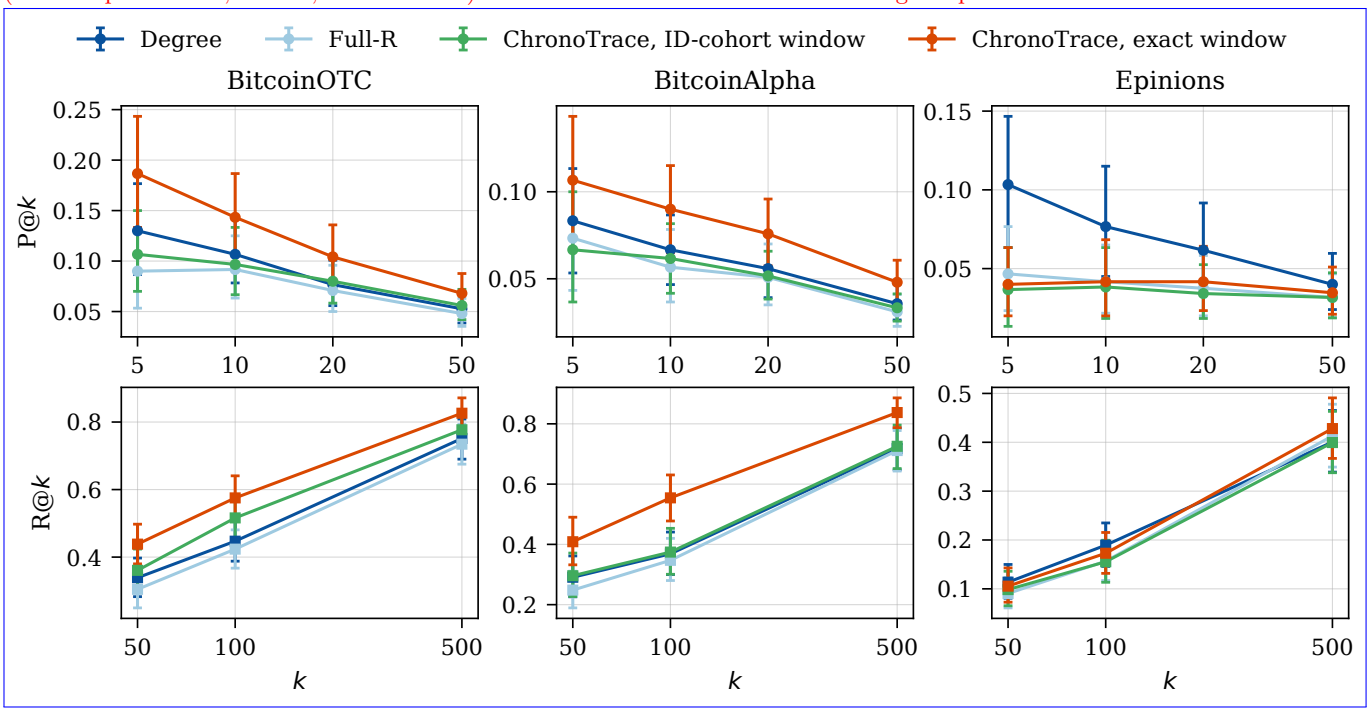


Figure 9: Precision (top) and recall (bottom) at depth k (95% CI over 60 targets).

Table 13: Paired differences in AUC against the Degree adversary (last row of each block: against Full-R), with 95% bootstrap CIs over 60 targets, two-sided sign-flip p-values, and Benjamini–Hochberg adjusted p-values within each dataset.

Dataset	Comparison	ΔAUC	95% CI	p	p_{BH}
BitcoinOTC	Topology	−0.002	[−0.006, +0.001]	0.235	0.235
	Full-R	−0.010	[−0.019, −0.002]	0.025	0.029
	ChronoTrace, hub window	−0.011	[−0.021, −0.002]	0.021	0.029
	ChronoTrace, ID-cohort window	+0.018	[+0.003, +0.031]	0.010	0.016
	Temporal	+0.029	[+0.014, +0.044]	<0.001	<0.001
	Sgn-Struct	+0.036	[+0.022, +0.050]	<0.001	<0.001
	ChronoTrace, exact window	+0.039	[+0.026, +0.053]	<0.001	<0.001
	ChronoTrace (exact) minus Full-R	+0.048	[+0.034, +0.064]	<0.001	<0.001
BitcoinAlpha	Topology	+0.002	[−0.006, +0.010]	0.630	0.720
	Full-R	−0.001	[−0.010, +0.008]	0.776	0.776
	ChronoTrace, hub window	−0.013	[−0.033, +0.005]	0.212	0.339
	ChronoTrace, ID-cohort window	+0.011	[−0.010, +0.033]	0.324	0.432
	Temporal	+0.038	[+0.014, +0.062]	0.002	0.004
	Sgn-Struct	+0.061	[+0.043, +0.081]	<0.001	<0.001
	ChronoTrace, exact window	+0.062	[+0.043, +0.083]	<0.001	<0.001
	ChronoTrace (exact) minus Full-R	+0.063	[+0.046, +0.081]	<0.001	<0.001
Epinions	Topology	+0.002	[−0.003, +0.006]	0.457	0.915
	Full-R	−0.002	[−0.007, +0.003]	0.438	0.915
	ChronoTrace, hub window	−0.005	[−0.013, +0.002]	0.223	0.892
	ChronoTrace, ID-cohort window	+0.002	[−0.006, +0.009]	0.656	0.958
	Temporal	−0.021	[−0.037, −0.006]	0.008	0.066
	Sgn-Struct	+0.002	[−0.008, +0.012]	0.728	0.958
	ChronoTrace, exact window	−0.001	[−0.017, +0.012]	0.861	0.958
	ChronoTrace (exact) minus Full-R	0.000	[−0.014, +0.013]	0.958	0.958

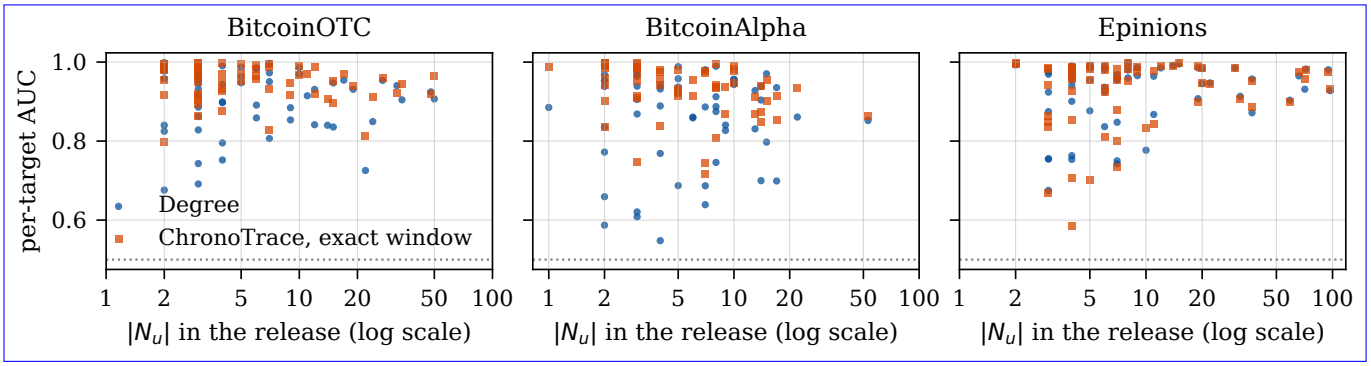


Figure 10: Per-target AUC of the Degree and ChronoTrace (exact window) adversaries against $|N_u|$.

D.6 Sign and time of deleted relationships

Table 17 gives the numbers behind Section 6.4: for every dataset, the ~~same distribution visible to the attacker)while destroying actual adjacency information. We run 200 Monte Carlo trials per task and report the mean AUC across all trials and tasks.~~ share of pairs whose relationship included distrust, the AUC of the sign predictor with its confidence interval, and the accuracy of every time predictor of Appendix B.7.

~~The calibration values themselves are reported in~~

D.7 Rare former neighbors

Table 7 groups former neighbors by degree. Table 18 summarizes the same question in one number per adversary, the ~~main body (Table ??), because the gap between the actual and null AUC is the paper's primary quantitative claim. Two further observations belong here. On BitcoinAlpha and Epinions, where the actual degreeAUC falls below the null, deleted nodes'neighbors are evidently not unusually high-degree; some are low-degree peripheral nodes whose removal leaves little structural residue. The Epinions case is the extreme: the degree-proportional null (0.912) nearly matches the full chronotrace attacker (0.928), so on a large dense graph structural prominence alone accounts for almost all observable signal~~surprisal-weighted recall, in which rare former neighbors count more. It weights each former neighbor v by $-\log q_v$, where $q_v = (\deg v + 1)/\sum_{w \in G_u} (\deg w + 1)$. The weighting lowers every adversary's recall, because rare neighbors are harder to find. On the Bitcoin graphs it costs ChronoTrace with the exact window less than Degree: at depth 100, Degree loses 14% and 14% of its recall, and ChronoTrace 10% and 7%.

D.8 ~~PA/Degree Prior Equivalence~~

~~Preferential attachment (PA) and degree~~

Table 14: ChronoTrace under each level of timing knowledge (mean over 60 targets). IoU: overlap with the exact window. Δ: paired difference to the Degree adversary. Excess: excess over degree.

Dataset	Window	IoU	AUC [95% CI]	Δ vs Degree [95% CI]	DM-AUC	Excess
BitcoinOTC	exact	1.00	0.949 [0.937, 0.960]	+0.039 [+0.026, +0.053]	0.81	+0.079*
	shifted, $\delta = 1$	0.66	0.948 [0.937, 0.958]	+0.038 [+0.024, +0.052]	0.80	+0.077*
	shifted, $\delta = 2$	0.43	0.911 [0.889, 0.932]	+0.001 [−0.014, +0.015]	0.63	+0.026*
	shifted, $\delta = 4$	0.22	0.906 [0.881, 0.929]	−0.004 [−0.021, +0.011]	0.62	+0.020*
	shifted, $\delta = 8$	0.12	0.893 [0.870, 0.915]	−0.017 [−0.029, −0.006]	0.49	−0.003
	ID cohort	0.66	0.928 [0.911, 0.944]	+0.018 [+0.003, +0.031]	0.70	+0.040*
	hub	0.26	0.899 [0.875, 0.921]	−0.011 [−0.021, −0.002]	0.55	+0.002
BitcoinAlpha	exact	1.00	0.931 [0.914, 0.947]	+0.062 [+0.043, +0.083]	0.81	+0.109*
	shifted, $\delta = 1$	0.68	0.931 [0.913, 0.946]	+0.061 [+0.042, +0.082]	0.81	+0.107*
	shifted, $\delta = 2$	0.46	0.890 [0.862, 0.916]	+0.021 [+0.009, +0.035]	0.72	+0.051*
	shifted, $\delta = 4$	0.30	0.881 [0.853, 0.907]	+0.011 [−0.006, +0.027]	0.68	+0.045*
	shifted, $\delta = 8$	0.16	0.861 [0.832, 0.889]	−0.008 [−0.017, +0.002]	0.50	+0.004
	ID cohort	0.53	0.880 [0.847, 0.909]	+0.011 [−0.010, +0.033]	0.57	+0.040*
	hub	0.22	0.856 [0.822, 0.888]	−0.013 [−0.033, +0.005]	0.61	+0.007
Epinions	exact	1.00	0.923 [0.899, 0.944]	−0.001 [−0.017, +0.012]	0.59	+0.011
	shifted, $\delta = 1$	0.80	0.924 [0.901, 0.945]	0.000 [−0.016, +0.013]	0.61	+0.012
	shifted, $\delta = 2$	0.67	0.921 [0.898, 0.941]	−0.003 [−0.019, +0.009]	0.55	+0.007
	shifted, $\delta = 4$	0.53	0.915 [0.890, 0.938]	−0.009 [−0.024, +0.001]	0.49	0.000
	shifted, $\delta = 8$	0.47	0.920 [0.898, 0.940]	−0.004 [−0.011, +0.004]	0.44	+0.005
	ID cohort	0.48	0.926 [0.905, 0.945]	+0.002 [−0.006, +0.009]	0.59	+0.011*
	hub	0.22	0.919 [0.898, 0.939]	−0.005 [−0.013, +0.002]	0.48	+0.001

* $p < 0.05$, two-sided sign-flip permutation test (unadjusted).

Table 15: AUC of the three window-based feature sets under each level of timing knowledge (mean over 60 targets); first row: the Degree adversary.

	BitcoinOTC			BitcoinAlpha			Epinions		
Window	Temporal	Sgn-Struct	ChronoTrace	Temporal	Sgn-Struct	ChronoTrace	Temporal	Sgn-Struct	ChronoTrace
none (Degree)		0.910			0.869			0.924	
exact	0.939	0.946	0.949	0.907	0.930	0.931	0.903	0.926	0.923
shifted, $\delta = 1$	0.939	0.944	0.948	0.908	0.930	0.931	0.907	0.928	0.924
shifted, $\delta = 2$	0.914	0.905	0.911	0.879	0.876	0.890	0.904	0.924	0.921
shifted, $\delta = 4$	0.908	0.900	0.906	0.874	0.865	0.881	0.901	0.917	0.915
shifted, $\delta = 8$	0.847	0.897	0.893	0.773	0.860	0.861	0.888	0.920	0.920
ID cohort	0.921	0.927	0.928	0.837	0.880	0.880	0.904	0.925	0.926
hub	0.862	0.902	0.899	0.787	0.861	0.856	0.886	0.921	0.919

Table 16: Targets with one to three events (60 drawn per dataset). Best: mean rank of the best-ranked former neighbor. n: targets with at least one former neighbor in the edge-list release.

Dataset	Feature set	AUC [95% CI]	S@10	Best	n
BitcoinOTC	Degree	0.953 [0.925, 0.975]	0.32	187	59
	Full-R	0.954 [0.928, 0.974]	0.29	198	59
	ChronoTrace, ID-cohort window	0.971 [0.954, 0.984]	0.31	121	59
	ChronoTrace, exact window	0.978 [0.965, 0.988]	0.36	99	59
BitcoinAlpha	Degree	0.893 [0.845, 0.932]	0.27	333	60
	Full-R	0.901 [0.857, 0.937]	0.20	317	60
	ChronoTrace, ID-cohort window	0.913 [0.875, 0.944]	0.25	273	60
	ChronoTrace, exact window	0.956 [0.938, 0.972]	0.23	138	60
Epinions	Degree	0.903 [0.863, 0.935]	0.04	4316	46
	Full-R	0.898 [0.849, 0.937]	0.09	4401	46
	ChronoTrace, ID-cohort window	0.901 [0.848, 0.940]	0.09	4311	46
	ChronoTrace, exact window	0.906 [0.859, 0.942]	0.09	3677	46

Table 17: Sign and time of deleted relationships, over pairs of a main target and a former neighbor present in the edge-list release. Distrust: share of pairs with a negative rating. Mode: the dataset's busiest bin; Busiest: the neighbor's busiest bin; W_u: exact window.

				Same bin				Within one bin			
Dataset	Pairs	Distrust	Sign AUC [95% CI]	Mode	Busiest	Busiest in W_u	Middle of W_u	Mode	Busiest	Busiest in W_u	Middle of W_u
BitcoinOTC	543	14%	0.82 [0.76, 0.88]	9%	43%	50%	29%	14%	61%	75%	56%
BitcoinAlpha	443	5%	0.82 [0.72, 0.92]	26%	47%	50%	29%	30%	69%	80%	50%
Epinions	968	23%	0.86 [0.83, 0.89]	4%	31%	36%	16%	19%	47%	62%	31%

Table 18: Recall (R) and surprisal-weighted recall (SWR) at depths 100 and 500 (mean over 60 targets).

Dataset	Feature set	Window	R@100	SWR@100	R@500	SWR@500
BitcoinOTC	Degree	none	0.447	0.384	0.751	0.705
	Full-R	none	0.424	0.360	0.734	0.686
	ChronoTrace	ID cohort	0.516	0.452	0.777	0.735
	ChronoTrace	exact	0.575	0.517	0.826	0.791
BitcoinAlpha	Degree	none	0.369	0.318	0.718	0.680
	Full-R	none	0.347	0.300	0.711	0.672
	ChronoTrace	ID cohort	0.374	0.333	0.726	0.694
	ChronoTrace	exact	0.554	0.514	0.837	0.812
Epinions	Degree	none	0.189	0.150	0.401	0.343
	Full-R	none	0.157	0.127	0.413	0.355
	ChronoTrace	ID cohort	0.155	0.124	0.399	0.345
	ChronoTrace	exact	0.173	0.140	0.428	0.370

D.8 Adapted heuristics

Figure 11 plots the heuristics of Table 9 with their confidence intervals. Preferential attachment and the neighbor-count prior produce identical ~~AUC, AP, and Recall values on every task for both datasets. This is a mathematical consequence: PA is computed as $\deg(v) \cdot \bar{d}_A$, where $\bar{d}_A$ is the mean degree of the oracle window anchor set. Since $\bar{d}_A$ is constant across all candidates within a single task, PA is a positive scalar multiple of $\deg(v)$. Degree prior is $\log(1 + \deg(v))$, which is monotonic in $\deg(v)$. Both therefore induce the same ranking. PA is retained in Table 9 for completeness and to demonstrate that the anchor set construction does not contribute independent signal beyond degree in this setup.~~ rankings: preferential attachment is $|N(v)| \cdot \bar{n}_A$, where the anchors' mean neighbor count $\bar{n}_A$ is constant within a target, and the prior $\log(1 + |N(v)|)$ is monotone in $|N(v)|$. Both rank by neighbor count, whereas the Degree adversary ranks by degree, the number of events; the two differ when ratings are reciprocated, which explains their slightly different results.

D.9 ~~Epinions Candidate Subsampling Sensitivity~~

~~The Epinions dataset (131, 828 nodes) requires candidate subsampling to 5, 000 per task due to computational constraints under the 5-seed protocol. To verify that subsampling does not materially affect the results, we conducted a sensitivity analysis on 12 Epinions tasks (seed 42)at three candidate pool sizes: 2, 000, 5,000 (default), and 10, 000. Table ?? reports the results.~~

D.9 Classifier family

A random forest (100 trees, depth 10, balanced class weights) trained on the same simulated erasures does not rank better than the logistic adversary for any feature set (Table 19). On Degree alone it ranks much worse, so it is a weaker attacker than the logistic regression rather than a stronger one.

~~AUC varies by less than 0.006 across the tested range, confirming that the subsampling does not materially affect the ranking metric. The slight decrease at larger pool sizes is expected: adding more negative candidates marginally dilutes the positive signal. Importantly, all true positives are preserved in every subsampling condition; only the number of negative candidatesvaries. The degree-proportional null model calibration is similarly stable (null AUC of 0.914 at 2, 000, 0.912 at 5, 000, and 0.911 at 10, 000). We acknowledge that evaluating the full 131, 828-node universe would provide a definitive comparison, but the computational cost under 5 seeds × 12 nodes × full feature extraction is prohibitive. The sensitivity analysis supports the conclusion that Epinions results are comparable to the full-universe Bitcoin evaluations for the purpose of cross-dataset comparison.~~

D.10 ~~Binary Trust Sign Sensitivity~~

D.10 Sign binarization

~~The core benchmark maps graph ratings > 0 to positive signs and ≤ 0 to negative signs. To verify that this zero-threshold binarization does not introduce artificial sensitivity, we evaluated the attacker across a range of binarization thresholds $\{-1, 0, 1, 5\}$ on BitcoinOTC and BitcoinAlpha (where raw ratings span~~ Bitcoin ratings range from -10 to $+10$ ~~). The results show negligible variation: on BitcoinOTC, AUC is 0.936 ± 0.054 at threshold -1 and 0,~~ and are never zero, so the thresholds $\tau = -1$ and ~~0.935 ± 0.067 at thresholds 1 and 5. On BitcoinAlpha, AUC is 0.922 ± 0.075 at threshold -1 and 0, and 0.917 ± 0.078 at threshold 1. This confirms the benchmark's qualitative findings are insensitive to the exact rating threshold used for binary signs.~~ $\tau = 0$ give identical signs. Table 20 varies the threshold above which a rating counts as positive; ChronoTrace's AUC barely changes on either Bitcoin graph. Epinions statements carry only a sign.

D.11 Temporal trust GNN embeddings

DGTEN [29] is a temporal trust-prediction graph neural network. We use a deterministic variant without its uncertainty modeling: three signed graph-convolution layers followed by a temporal attention module. It is trained for 50 epochs on the first seven of ten equal-length time snapshots of each Bitcoin graph before erasure, with input features smoothed over the whole graph, and we append its 32-dimensional node embeddings to the candidate features. The embeddings are learned from the graph before erasure, which contains the deleted edges, so this baseline sees more than a release-only adversary may. Even so, the embeddings rank former neighbors worse than degree, and adding them to the hand-engineered features does not help (Table 21). The embeddings are optimized for sign prediction, not for neighbor inference, so this result does not show that graph neural networks are unsuitable for the task.

D.12 Defenses in full

Table 22 adds standard deviations, recall at depth 500, and the spread of the share of events kept to Table 8. Sign prediction from endpoint features is weak on BitcoinAlpha, so its utility column says less than those of the other datasets.

Table 19: Logistic regression (LR) and random forest (RF) attackers trained on the same simulated erasures (mean AUC over 60 targets).

	Degree		Full-R		ChronoTrace (exact)	
Dataset	LR	RF	LR	RF	LR	RF
BitcoinOTC	0.910	0.756	0.901	0.866	0.949	0.929
BitcoinAlpha	0.869	0.725	0.868	0.838	0.931	0.903
Epinions	0.924	0.852	0.922	0.888	0.923	0.897

Table 20: ChronoTrace (exact window) when ratings above τ count as positive (mean AUC over 60 targets).

Dataset	$\tau = -1$	$\tau = 0$	$\tau = 1$	$\tau = 5$	Distrust at $\tau = 0$
BitcoinOTC	0.949	0.949	0.950	0.949	10%
BitcoinAlpha	0.931	0.931	0.928	0.929	6%

Table 21: Node embeddings of a DGTEN variant as attacker features (mean±std over 60 targets; Δ: paired difference to Degree with 95% CI).

Dataset	Features	AUC	Δ vs Degree
BitcoinOTC	Degree	0.910±0.081	–
	DGTEN embedding	0.833±0.133	−0.077 [−0.107, −0.051]
	Full-R + DGTEN	0.869±0.122	−0.042 [−0.070, −0.021]
	ChronoTrace (exact)	0.949±0.047	+0.039 [+0.026, +0.053]
	ChronoTrace (exact) + DGTEN	0.931±0.076	+0.020 [−0.001, +0.040]
BitcoinAlpha	Degree	0.869±0.119	–
	DGTEN embedding	0.752±0.199	−0.117 [−0.157, −0.078]
	Full-R + DGTEN	0.800±0.171	−0.069 [−0.103, −0.039]
	ChronoTrace (exact)	0.931±0.066	+0.062 [+0.043, +0.083]
	ChronoTrace (exact) + DGTEN	0.887±0.106	+0.018 [−0.010, +0.046]

Table 22: Defenses in full (mean±std over 60 targets): AUC of both adversaries, R@500 of ChronoTrace, share of events kept, and sign-prediction utility. RR: randomized response.

Dataset	Defense	Degree AUC	ChronoTrace AUC	ChronoTrace R@500	Kept	Utility
BitcoinOTC	None	0.910±0.081	0.949±0.047	0.826	1.00±0.00	0.635±0.012
	ChronoScrub, temporal	0.875±0.117	0.933±0.070	0.792	0.91±0.07	0.632±0.016
	ChronoScrub, joint 0.25	0.901±0.091	0.943±0.057	0.802	0.97±0.02	0.636±0.014
	ChronoScrub, joint 0.50	0.895±0.097	0.942±0.057	0.793	0.94±0.04	0.632±0.013
	ChronoScrub, joint 0.75	0.886±0.111	0.939±0.061	0.823	0.92±0.06	0.633±0.011
	RR, $\epsilon = 0.5$	0.904±0.087	0.944±0.052	0.813	0.62±0.00	0.639±0.014
	RR, $\epsilon = 1$	0.907±0.086	0.946±0.050	0.826	0.73±0.00	0.637±0.012
	RR, $\epsilon = 2$	0.910±0.080	0.948±0.048	0.810	0.88±0.00	0.637±0.013
	RR, $\epsilon = 4$	0.910±0.081	0.949±0.046	0.813	0.98±0.00	0.639±0.018
BitcoinAlpha	None	0.869±0.119	0.931±0.066	0.837	1.00±0.00	0.545±0.012
	ChronoScrub, temporal	0.825±0.144	0.928±0.070	0.829	0.91±0.09	0.544±0.011
	ChronoScrub, joint 0.25	0.860±0.125	0.929±0.069	0.842	0.97±0.02	0.545±0.011
	ChronoScrub, joint 0.50	0.846±0.137	0.932±0.069	0.837	0.94±0.05	0.549±0.011
	ChronoScrub, joint 0.75	0.836±0.137	0.934±0.070	0.866	0.92±0.07	0.550±0.010
	RR, $\epsilon = 0.5$	0.856±0.130	0.925±0.075	0.854	0.62±0.00	0.544±0.011
	RR, $\epsilon = 1$	0.861±0.127	0.926±0.078	0.856	0.73±0.00	0.544±0.011
	RR, $\epsilon = 2$	0.867±0.120	0.930±0.074	0.840	0.88±0.00	0.547±0.011
	RR, $\epsilon = 4$	0.869±0.119	0.930±0.067	0.836	0.98±0.00	0.548±0.010
Epinions	None	0.924±0.081	0.923±0.092	0.428	1.00±0.00	0.767±0.013
	ChronoScrub, temporal	0.920±0.081	0.918±0.091	0.387	0.86±0.11	0.768±0.014
	ChronoScrub, joint 0.25	0.928±0.077	0.926±0.091	0.425	0.96±0.03	0.766±0.012
	ChronoScrub, joint 0.50	0.928±0.077	0.926±0.091	0.404	0.92±0.05	0.766±0.013
	ChronoScrub, joint 0.75	0.920±0.083	0.921±0.087	0.381	0.88±0.08	0.767±0.013
	RR, $\epsilon = 0.5$	0.919±0.081	0.915±0.095	0.434	0.62±0.00	0.767±0.014
	RR, $\epsilon = 1$	0.927±0.075	0.924±0.087	0.437	0.73±0.00	0.768±0.014
	RR, $\epsilon = 2$	0.924±0.082	0.925±0.088	0.459	0.88±0.00	0.764±0.015
	RR, $\epsilon = 4$	0.923±0.081	0.922±0.094	0.436	0.98±0.00	0.769±0.012

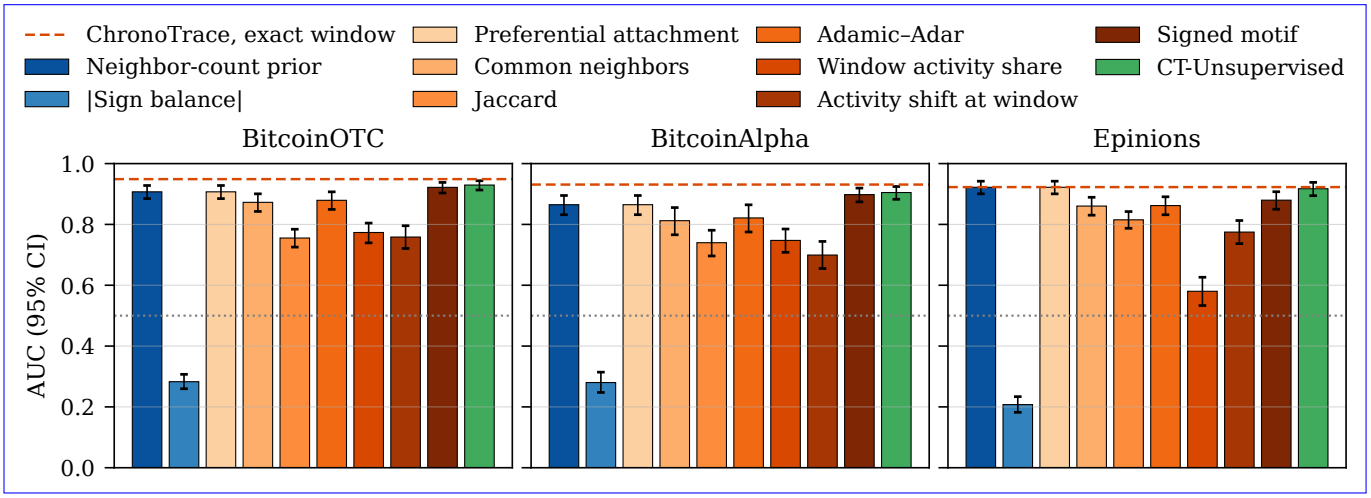


Figure 11: Adapted heuristics (AUC, 95% CI). Dashed: ChronoTrace with the exact window.

E Continuity with the Submitted Evaluation

Besides its training protocol (Appendix C.2), the submitted version differed from this revision in three ways: it evaluated directory releases, it subsampled 5,000 candidates on BitcoinOTC and Epinions, and it used the static SNAP Epinions snapshot. This appendix reports each of the three settings.

Directory releases. Table 23 reports the main adversaries in the directory setting. The isolated former neighbors lower the Degree adversary there. They produce the negative gaps to the degree reference that the submitted version reported and called deletion leakage, and the gains of structural features that it reported on BitcoinAlpha and Epinions.

Candidate pools. Table 24 and Figure 12 show how the ranking metrics depend on the size of the candidate pool. AUC does not change with the pool size, whereas AP, P@5, S@10 and the best rank degrade as more non-neighbors compete for the top positions.

Static Epinions snapshot. Table 25 evaluates the static snapshot on its full universe of 131,827 candidates. Its time column is the record order, and 97% of consecutive records have a rater ID at least as large as the previous record's, so the order largely follows rater IDs. The timing features therefore read file order, and ChronoTrace's gain of +0.014 (95% CI [+0.003, +0.026]) over Degree does not measure timing knowledge. Unlike on the three dated graphs, structure ranks candidates of equal degree slightly above chance on this snapshot: Topology has degree-matched AUC 0.54 and excess over degree +0.007, although its AUC gain over Degree, +0.003 (95% CI [−0.001, +0.006]), is not significant. Full-R ranks candidates of equal degree better still (degree-matched AUC 0.60), because its activity concentration (feature 10) reads record order.

F Reproducibility

Every number in this paper is a macro, and every data table and plot is a file, generated by two scripts from the per-target result files; no number is typed by hand. The released artifact (see Open Science) contains the code, every per-target record, the run logs and these scripts. A preparation script rebuilds the input files from the public releases and checks them against SHA-256 checksums of the files we used, and it reproduces all of them byte for byte. Every random choice is seeded: target sampling, negative sampling, window shifts, rewiring, label references, randomized response, the bootstrap and the permutation tests. With one BLAS thread per worker, rerunning the experiments reproduces the saved records; we checked this by rerunning one seed of BitcoinOTC from scratch, which gave the same values for every metric, coefficient and window of its 12 targets.

All experiments ran with Python 3.10, NumPy 1.26, SciPy 1.15, scikit-learn 1.6, pandas 2.3 and Matplotlib 3.10; the GNN baseline additionally used PyTorch 2.4. Regenerating every table and figure from the saved records takes about two minutes and needs neither the datasets nor a GPU. The artifact is organized as follows:

- `code/chronotrace/`: the library (features, adversaries and training, window estimators, reference graphs, degree controls, defenses);
- `code/scripts/`: data preparation, experiment runners, analyses and the two generator scripts;
- `data/`: download instructions, checksums and the provenance of dated Epinions (no datasets);
- `paper/results/revision/`: one JSON record per target and experiment, aggregated summaries with bootstrap intervals and tests, and batch logs;
- `tests/`: unit tests on synthetic graphs and end-to-end tests on BitcoinOTC.

Table 23: The submitted evaluation setting: directory releases, in which users left without events remain candidates (mean over 60 targets).

Dataset	Feature set	AUC [95% CI]	AP	P@5	DM-AUC
BitcoinOTC	Degree	0.895 [0.870, 0.919]	0.110	0.13	0.50
	Topology	0.899 [0.874, 0.921]	0.087	0.10	0.49
	Full-R	0.899 [0.875, 0.921]	0.081	0.11	0.52
	Temporal	0.942 [0.930, 0.953]	0.103	0.14	0.78
	Sgn-Struct	0.935 [0.917, 0.950]	0.117	0.12	0.73
	ChronoTrace	0.947 [0.934, 0.958]	0.122	0.18	0.81
	Degree minus label reference	+0.040 [+0.013, +0.064]			
BitcoinAlpha	Degree	0.838 [0.800, 0.873]	0.086	0.08	0.50
	Topology	0.871 [0.843, 0.897]	0.068	0.06	0.54
	Full-R	0.869 [0.842, 0.896]	0.067	0.07	0.55
	Temporal	0.908 [0.885, 0.929]	0.094	0.10	0.80
	Sgn-Struct	0.925 [0.907, 0.941]	0.105	0.12	0.80
	ChronoTrace	0.930 [0.914, 0.946]	0.103	0.11	0.82
	Degree minus label reference	−0.015 [−0.053, +0.021]			
Epinions	Degree	0.830 [0.784, 0.873]	0.048	0.10	0.50
	Topology	0.920 [0.902, 0.936]	0.020	0.02	0.58
	Full-R	0.928 [0.910, 0.943]	0.021	0.04	0.51
	Temporal	0.915 [0.895, 0.934]	0.022	0.04	0.61
	Sgn-Struct	0.915 [0.896, 0.934]	0.019	0.01	0.62
	ChronoTrace	0.927 [0.907, 0.944]	0.021	0.04	0.59
	Degree minus label reference	−0.083 [−0.130, −0.039]			

Table 24: Degree adversary as the candidate pool shrinks (every former neighbor kept; mean over targets).

Dataset	Pool	AUC	AP	P@5	S@10	Best
Epinions (dated)	2,000	0.924	0.275	0.34	0.81	7
	5,000	0.924	0.182	0.25	0.70	16
	10,000	0.924	0.128	0.20	0.57	30
	20,000	0.924	0.090	0.16	0.50	59
	50,000	0.924	0.055	0.11	0.38	147
	all	0.924	0.051	0.10	0.35	167
Epinions (static)	2,000	0.914	0.251	0.33	0.81	15
	5,000	0.914	0.162	0.22	0.72	36
	10,000	0.914	0.111	0.16	0.62	70
	20,000	0.914	0.070	0.11	0.49	140
	50,000	0.914	0.037	0.06	0.33	348
	all	0.914	0.017	0.02	0.22	930

Table 25: Static SNAP Epinions, edge-list release, full candidate universe (60 targets). DM-AUC: degree-matched AUC. Excess: excess over degree.

Feature set	AUC [95% CI]	AP	P@5	DM-AUC	Excess
Degree	0.914 [0.893, 0.934]	0.017	0.02	0.50	0.000
Topology	0.917 [0.896, 0.936]	0.019	0.04	0.54	+0.007*
Full-R	0.917 [0.897, 0.936]	0.022	0.04	0.60	+0.013*
ChronoTrace	0.929 [0.910, 0.946]	0.027	0.04	0.71	+0.038*

* $p < 0.05$, two-sided sign-flip permutation test (unadjusted).

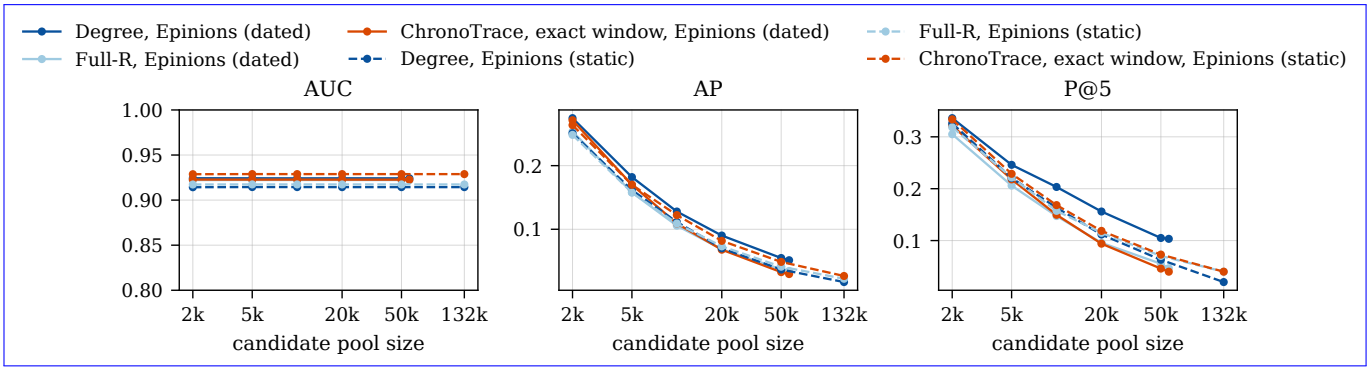


Figure 12: AUC, AP and P@5 as the candidate pool shrinks, with every former neighbor kept.